**A CENTRALIZED SYSTEM FOR REAL-TIME FARM ANIMAL DISEASE REPORTING**

# Common Integrated Solution Report

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**Abstract**

Livestock health information is often distributed across farm observations, diagnostic evidence and administrative surveillance. This research develops the Animal Disease Reporting System (ADRS), a veterinary-supervised framework integrating persistent wellness monitoring and body condition scoring, cattle-image assessment, geospatial surveillance and climate-informed district-month outbreak-risk estimation. The shared architecture uses a React frontend, FastAPI services and MongoDB operational storage, with distinct model and historical-data artefacts. Wellness monitoring reported 99.80% accuracy on engineered data; a base late-fusion model achieved complete agreement on 20 simulated profiles, which does not establish clinical generalisation. The image component achieved 90.11% ViT-B/16 test accuracy and improved segmentation mean Intersection-over-Union from approximately 0.298 to 0.4078 on 100 samples. YOLOv8s validation curves indicated approximately 0.81 mAP@50 and 0.54 mAP@50–95. Geospatial outputs connect verified clinical records with historical source-aware relative spatial scores and geometric descriptors. The enhanced FMD model achieved held-out 2024 ROC-AUC 0.8166 and PR-AUC 0.6968, while the calibrated LSD model achieved five-fold Leave-One-Year-Out mean ROC-AUC 0.6136 and PR-AUC 0.2699. Future-period estimates use disclosed historical climate proxies where observations are unavailable. Selected forecasting integration workflows and an automated checkpoint reporting 862 passes support application feasibility. However, complete image-service orchestration, prospective clinical validation, system-wide security and recovery testing, and representative stakeholder acceptance remain outstanding. The principal contribution is a common evidence framework connecting animal-level observations to veterinary review and population-level surveillance without treating model predictions as confirmed disease. The results support further supervised evaluation rather than autonomous diagnosis or production readiness.

Keywords: Livestock disease surveillance, multimodal triage, computer vision, geospatial analysis, outbreak-risk nowcasting

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# LIST OF ABBREVIATIONS

| Abbreviation | Description |
|---|---|
| ADRS | Animal Disease Reporting System |
| AI | Artificial Intelligence |
| API | Application Programming Interface |
| BCS | Body Condition Score |
| CHIRPS | Climate Hazards Infrared Precipitation with Stations |
| CNN | Convolutional Neural Network |
| DAPH | Department of Animal Production and Health |
| DSR | Design Science Research |
| EMPRES-i | Emergency Prevention System Global Animal Disease Information System |
| ERA5 | ECMWF fifth-generation atmospheric reanalysis |
| FAO | Food and Agriculture Organization of the United Nations |
| FMD | Foot-and-Mouth Disease |
| GO | Group Objective |
| GPS | Global Positioning System |
| Grad-CAM | Gradient-weighted Class Activation Mapping |
| HTTP | Hypertext Transfer Protocol |
| IoU | Intersection-over-Union |

| | |
|---|---|
| JWT | JSON Web Token |
| LOYO | Leave-One-Year-Out |
| LSD | Lumpy Skin Disease |
| LSTM | Long Short-Term Memory |
| MAE | Mean Absolute Error |
| mAP | Mean Average Precision |
| mIoU | Mean Intersection-over-Union |
| OOD | Out-of-Distribution |
| PR-AUC | Area Under the Precision–Recall Curve |
| RMSE | Root Mean Square Error |
| ROC-AUC | Area Under the Receiver Operating Characteristic Curve |
| ROI | Region of Interest |
| SDG | Sustainable Development Goal |
| THI | Temperature-Humidity Index |
| ViT | Vision Transformer |
| WAHIS | World Animal Health Information System |
| WCAG | Web Content Accessibility Guidelines |
| WGS84 | World Geodetic System 1984 |
| WOAH | World Organisation for Animal Health |

# 2. INTRODUCTION AND PROJECT CONTEXT

## 2.1 Background and Context

Livestock production contributes to food availability and to the livelihoods of people in rural areas in Sri Lanka. According to the Department of Animal Production and Health (DAPH), livestock represented about 1.4% of the gross domestic product at current prices in 2024. Milk production within the country went up by 5.6% that year, and the amount of milk per person annually increased from 39.98 liters to 40.30 liters. Small-scale production still forms a major portion of the sector [1] . Since animal health is important, it has an effect on both household income and on the continued operation of livestock production.

Foot-and-mouth disease (FMD) is a highly contagious disease of cloven-hoofed animals that can have an effect on production and on trade [2] . Lumpy skin disease (LSD) affects cattle and is linked to production losses and is mainly transmitted in a vector-mediated way [3] . In the case of these two diseases, a disease-specific interpretation is required since neither of them is a general sign of all livestock health problems. The experiments referred to deal with animal-health outcomes and do not involve the assessment of any risk to humans.

Sri Lanka has a well-established veterinary surveillance network, consisting of passive surveillance carried out by 337 divisional government veterinary offices, the major livestock diseases being assessed on the basis of clinical signs and reports being submitted on a monthly basis. Veterinary Investigation Centres provide support for both field and laboratory investigations [4] . This system continues to be essential and the project aims at improving the organization and interpretation of the information available to veterinary personnel within the system.

The Food and Agriculture Organization (FAO) offers early warning facilities through its disease intelligence, reporting and analysis services [5] . Likewise, the FAO's EMPRES-i and the World Organization for Animal Health (WOAH) World Animal Health Information System (WAHIS) also supply information on international disease events [6], [7] The fact that these systems exist shows that the practice of

digital reporting of animal diseases is already underway. The contribution of the ADRS lies in linking farm-level triage, image assessment, spatial context and district-month risk within a single Sri Lankan workflow.

Since practical design decisions are needed, a study of dairy farms in Italy found that barriers to the practice of precision livestock farming were the investment cost, the time required to look at the sensor data and the difficulty of interpreting the data [8] . The results suggest the development of an application that makes use of ordinary photographs and structured records, though it will be necessary to investigate the applicability of such an approach to farms in Sri Lanka on a local basis. A responsive interface lowers the requirements for installation; it does not eliminate the need for connectivity, training or good data quality.

The Animal Disease Reporting System (ADRS) consists of four complementary elements: Persistent Health Anomaly and Body Condition Scoring, AI Smart Diagnosis, Real-Time Geospatial Clustering and Trajectory Mapping, and Climate-Informed Livestock Disease Forecasting. Its main aim is to provide support for veterinary decisions. The outputs from the models aid in observation, review and the prioritization of surveillance; however, a professional assessment is necessary before a suspected condition can be recorded as a confirmed clinical case.

### 2.2 Stakeholders and User Needs

The main stakeholders include livestock farmers, veterinary officers, the district veterinary authorities and DAPH officials. Farmers need simple daily logging, the ability to submit images, clear feedback on the animals' health and access to veterinary advice. Veterinary officers need the animal's history, photographic evidence, information about the farm and a clear way to tell the difference between a suggestion made by the model and a confirmed diagnosis. Those at the district and national levels need surveillance organised by geography and estimates which can aid in planning investigations.

The surveillance duties of DAPH form the institutional background against which these roles are set [4] . Although the proposed application roles take account of that background, the research does not show that they have been formally adopted by

DAPH. Other users are epidemiologists, laboratories and authorised researchers, all of whom require traceable historical records, definitions of the outcomes and information on data quality.

The level of detail in stakeholder requirements also varies: a farmer's daily record of milk yield refers to a single animal, an image prediction refers to a particular region in a photograph, a clinical event relates to a case that has been reviewed by a veterinarian, and a district-month probability refers to aggregated historical occurrences. The application has to retain these different meanings. Table 2.1 gives a summary of the main user needs and the evidence presented to each role.

**Table 2.1 Stakeholder needs within the integrated ADRS workflow**

| Stakeholder | Main need | Relevant information |
|---|---|---|
| Farmer | Record observations and seek guidance | Daily logs, triage results, BCS and advisory information |
| Veterinary officer | Review suspected conditions and district cases | Animal history, image outputs, verified cases and area context |
| District authority | Prioritise investigations | Confirmed-event maps and district-month risk estimates |
| DAPH official | Review broader surveillance and follow-ups | National risk views, decision records and data provenance |
| Research or laboratory user | Assess evidence quality | Source lineage, model version and verification status |

## 2.3 Alignment with the Sustainable Development Goals

ADRS has a prospective alignment with Sustainable Development Goal (SDG) 2, especially with respect to targets 2.3 and 2.4 which relate to small-scale producers and resilient agricultural production [9] If the policy is implemented and results in effective action, earlier identification of animal-health issues together with more organised veterinary information could help to advance these objectives. This study focuses on assessing the technical capabilities rather than on showing higher productivity or less hunger.

Target 3.d of the SDGs relates to early warning systems and the management of health risks [10] . ADRS makes an indirect contribution from the point of view of animal health surveillance. The diseases it covers are solely veterinary in nature and the experiments it has reported do not involve measuring health outcomes in humans. It is therefore necessary to evaluate any contribution to public health separately, rather than deducing it from the accuracy of the classifier or the features of the maps.

## 2.4 Purpose and Scope of the Common Report

This report treats ADRS as a single research and software-engineering project and outlines the common problem, the way it has been divided into four subproblems, the shared architecture, the data boundaries, the integration of the components, the evaluation process, the requirements for adoption and the future work. The individual theses contain detailed evidence concerning wellness triage, smart diagnosis and district-level forecasting. The geospatial work is explained via the integrated architecture and the spatial-analysis account set out in this report.

The report separates completed experiments from demonstrated application workflows and from the requirements for wider deployment. It is important to make this distinction since a successful model experiment does not prove clinical validity, a recorded test execution does not certify any subsequent integration state, and a proposed deployment plan does not mean that production operations have been established. The key conclusion is that the individual component results do support a coherent decision-support framework, even though the overall system's effectiveness in real-world use has still got to be demonstrated.

# 3. OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

## 3.1 Overall Problem Definition

A major issue is that there is no single workflow in the project context which links regular animal observations, assessments based on images, verified clinical events, information on outbreaks arranged by location, and estimates of risk at the district level. Although each of the various sources of information can be useful on its own, the fragmented nature of the records makes it difficult to trace a concern from the first observation on a farm through to veterinary review and interpretation of the surveillance data.

The result in practice is a break in evidence. A decrease in milk production might lead to concern without clarifying if the thermal conditions are relevant. A photograph might indicate that an animal has a disease without confirming the diagnosis. A confirmed case might stay just a marker of a location without having any historical spatial context. Although historical district reports may be used to estimate the level of risk they do not offer an explanation relating to individual animals. The research looks at the relationships between these stages while keeping the different evidential status of each output.

## 3.2 Root Causes and Overall Research Gap

The initial reason is that there is limited contextual data available at the farm level. To interpret daily productivity and changes in body weight, it is necessary to take into account the breed of the animal, the stage of lactation, and the environmental conditions. Although temperature-humidity indices can be used to characterise thermal stress, their appropriateness depends on the type of housing, the local climate, and the characteristics of the animals [11]. A general heat threshold should not be regarded as a definite diagnosis of disease. For this reason, the triage component incorporates temporal physiological data, visual assessments of body condition score and environmental context.

The other reason is the narrow range of image-only inference. The public datasets of livestock that are used in computer vision have deficiencies in terms of environmental variety and contextual metadata [12]. A classifier might do well on the part of the data that was set aside for testing but could still fail when used on different farms, different lighting conditions, different species or in unfamiliar situations. Detecting out-of-distribution (OOD) cases is a separate area of research [13], and probability calibration needs its own evaluation [14]. Because of these factors, any conclusions drawn about the detector, classifier and segmenter that have been implemented should be treated with caution and it is not appropriate to claim that they can carry out autonomous diagnosis.

At the geographical level, the event identity, the quality of the coordinates and the time of observation all have an impact on what conclusions can be drawn from the reported outbreaks. Although ST-DBSCAN offers a method for clustering spatial and temporal observations [15], a cluster does not by itself constitute a transmission chain. ADRS separates the descriptive grouping of events, the retrospective spatial ranking and the cases that have been confirmed by veterinarians. The direction and reach displays are understood on the basis of how they have been calculated and not as reflecting actual pathogen movement.

The fourth reason is the scarcity and imbalance of the historical data on occurrence. The datasets for FMD and LSD, relating to district-monthly observations, include a relatively small number of positive cases. Precision-recall measures are used to evaluate the positive class [16], and when validating the results care must be taken to consider both temporal and spatial dependence [17] . Using a random split may lead to an incomplete understanding of the model's performance when it comes to later periods. Furthermore, future climate substitutions establish a difference between nowcasting based on observed conditions and projections made under climatological scenarios.

The issue in question is therefore not only one of integration and of having adequate evidence but also a modelling problem. In order for ADRS to function, it is necessary to have consistent identifiers, a defined boundary based on veterinary confirmation,

explicit output definitions and traceable provenance so that the information generated at one stage can be properly understood at the next stage. It is not necessary for all four of the components to use the same algorithm or numerical scale.

### 3.3 Subproblem and Component Mapping

Table 3.1 maps the four subproblems to their owners and common objectives. Component numbering is based on responsibility and remains consistent throughout this report. Component 1 is AI Smart Diagnosis; Component 2 is wellness triage; Component 3 is geospatial analysis; and Component 4 is district-level outbreak-risk assessment.

**Table 3.1 Subproblem and component mapping**

| Component and owner | Subproblem | Linked objective |
|---|---|---|
| C1 AI Smart Diagnosis — A.L.M. Athulathmudali, IT21129544 | Localise and classify cattle image regions and examine lesions | GO2 |
| C2 Persistent Health Anomaly and BCS — W.M.P.J. Wijenayake, IT22194558 | Interpret daily anomalies with body condition and thermal context | GO1 |
| C3 Geospatial Clustering and Trajectory Mapping — G.A. Sandaru, IT22258908 | Relate verified events to interpretable historical spatial context | GO3 |
| C4 Climate-Informed Forecasting — S.S. | Estimate district-month FMD and LSD occurrence risk | GO4 |

| | | |
|---|---|---|
| Kumarasinghe, IT22221414 | | |
| Shared application and integration | Connect records, roles and model outputs with traceable meaning | GO5 and GO6 |

The subproblems are dependent on one another but cannot be swapped. Triage raises a general concern regarding wellness; image analysis supplies visual evidence; clinical verification records the actual operational event; spatial analysis aids in interpreting the situation based on location; and district-month modelling contributes to a wider surveillance outlook. The common application links the various parts without permitting a risk score to take the place of clinical confirmation.

### 3.4 Overall Aim and Group Objectives

The aim is to develop and evaluate a framework for animal disease reporting and decision support which links observations at the farm level with image assessment, spatial surveillance and risk information at the district level in Sri Lanka.

The aim of GO1 is to carry out and evaluate persistent anomaly detection together with multimodal wellness triage by making use of productivity logs, BCS, metadata and thermal context. The evidence includes the engineered-data monitoring experiment, the comparison involving the 20 profiles, the payload processing and the visual explanation outputs. Separate acceptance criteria apply to clinical generalization.

GO2 will create an image processing workflow that is focused on cattle, enabling detection, four-class classification and optional lesion segmentation. The criteria that are measurable include the documentation of detector validation, the overall accuracy of the classifier in relation to the 85% research target, and an improvement in segmentation compared with the recorded baseline. The field robustness and end-to-end accuracy have to be evaluated separately.

GO3 is intended to offer information on confirmed cases organized by geography together with reproducible historical spatial analysis that includes the identified origins, sources, relative scores and interpretation limits. The evidence comprises the spatial processing and interface contracts, the veterinary district views and the recorded research demonstration.

GO4 is responsible for preparing and assessing the probability models for FMD and LSD across the 25 districts, using disease-specific temporal validation, ROC-AUC, PR-AUC and threshold analysis; the baseline and outbreak-history versions must each be reported separately and future proxy-based estimates must keep their provenance.

GO5 aims to create a common application framework featuring role-oriented interfaces, a shared identity and location context, validated requests, and a clear boundary between predictions and operational actions. For acceptance to be granted, there must be a traceable sequence going from submission through to review, confirmation, updating the map, and then follow-up, without any alteration of the model's semantics.

The purpose of GO6 is to assess the integrated artifact and to record information on its reliability, security, usability, evidence limits and deployment requirements. In order to achieve a full claim, it is necessary to carry out cross-component regression and failure-path testing, obtain acceptance from relevant stakeholders and establish a reproducible deployment checkpoint. Chapter 6 links the evidence that has been recorded to each of the six objectives.

## 3.5 Scope, Assumptions and Constraints

The farm and image sections focus on cattle. Wellness triage deals with the Healthy, Heat Stress and Unhealthy categories and does not identify a specific infectious disease. AI Smart Diagnosis makes use of the cattle, lumpy skin, foot-and-mouth and mastitis labels. The cattle label is an annotation category and should not be shown as indicating that the animal has been confirmed by a veterinarian as healthy. Cases of concurrent diseases and species transfer are outside the scope of the validated classifier.

The part dealing with forecasting includes data on FMD and LSD for Sri Lanka's 25 administrative districts. The processed FMD grid includes 2,400 district-months from 2017 to 2024, while the LSD grid has 1,500 entries from 2020 to 2024. These grids give information on reported cases only and do not provide a complete picture of infection prevalence or the outcomes for individual animals. Potential biases still exist due to missing reports and the inconsistent nature of surveillance.

The historical disease-source events are differentiated from the present verified clinical records by means of the geospatial component. The LSD reference score employed is based on cells of about 5 km and a 25 km distance-decay kernel. The score represents relative spatial context and is not a calibrated infection probability. The geometrical direction descriptor does not indicate independently observed transmission, and the nominal reach display is not a validated future outbreak boundary.

The system is based on the assumption that usable photographs are available, that the dates are accurate and daily logs are kept, that there is consistency in the district identifiers, that the model artefacts are accessible and that there is authorised veterinary review. Weather services and surveillance updates are considered external dependencies. If future observations are not available, historical data from the same calendar month can be used instead, and this substitution must be disclosed. Automatic treatment, quarantine, vaccination policy, restrictions on animal movements, unrestricted public access to farm locations and a national production deployment are not within the scope that has been demonstrated.

# 4. OVERALL SOLUTION STRUCTURE

## 4.1 Integrated Solution Overview

ADRS is organised around a shared animal-health record and a veterinary review workflow. A farmer contributes daily observations or initiates a scan. Wellness analysis can raise a concern and provide context; image assessment can provide a candidate label and optional visual evidence. A veterinary officer examines those records, adds clinical judgement and creates or verifies the operational case. Surveillance users then inspect the confirmed event geographically and consult district-level risk estimates as a separate planning aid.

This design avoids forcing every concern into a single automated pipeline. A farmer can request routine BCS assessment without submitting an infectious-disease case; a veterinarian can review a confirmed event without invoking an image classifier; and a DAPH-oriented user can request district-level risk without access to identifiable animal images. The contribution is a common information framework with controlled transitions, rather than an assertion that all four outputs measure the same underlying quantity.

## 4.2 High-Level Architecture

The system is built with a React front end, a FastAPI service layer, and MongoDB for data storage. Different services manage wellness inference, smart diagnosis, geospatial analysis, and district-month risk. Research models use historical surveillance and environmental data, while shared interfaces provide information about identity, farms, animals, and veterinary districts. Figure 4.1 shows these boundaries and highlights which application connections are complete and which, like the smart-diagnosis service interface, still need full runtime validation.

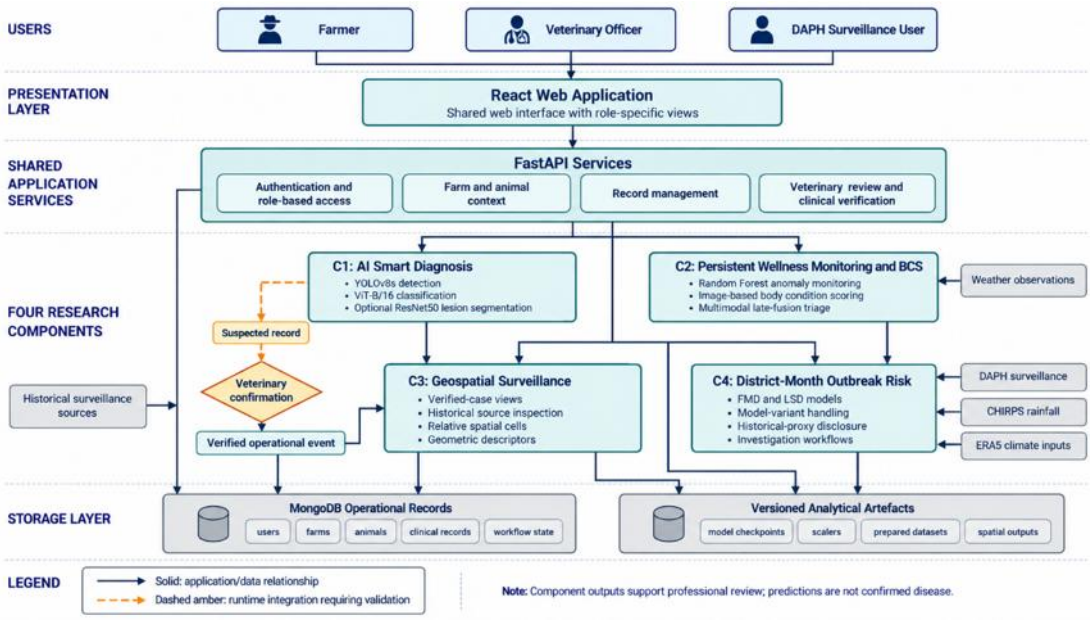



**Figure 4.1 High-level ADRS architecture and component boundaries**

Operational data and analytical outputs have different roles. MongoDB keeps application records and workflow states, while trained checkpoints, scalers, district-month files, and spatial research results need version control. Loading historical surveillance events into the research layer does not make them operational farm cases. Each service must identify the model or analysis used for its output.

The shared architecture allows components to reuse navigation, request handling, and directory context. However, it also means that issues like authentication problems, database outages, or inconsistent identifiers can impact multiple parts of the system. To manage this, each component keeps its own computations within defined services and shows clear unavailable states instead of pretending everything is normal or low-risk when a dependency fails.

## 4.3 Component 1 Summary

### 4.3.1 Owner and subproblem

AI Smart Diagnosis was developed by A.L.M. Athulathmudali (IT21129544). It addresses the use of ordinary cattle photographs to localize relevant regions, classify their visual appearance and optionally examine lesions. The implemented research artefact consists of YOLOv8s detection, ViT-B/16 classification and a ResNet50-based segmentation workflow. The broader guarded-diagnosis concept includes uncertainty and unfamiliar-input handling, but a validated OOD gate is not an established result.

### 4.3.2 Inputs and outputs

The input is a cattle photograph or an extracted region of interest (ROI). YOLOv8s produces bounding boxes, scores and detection labels. Crop coordinates are derived from image dimensions; square padding preserves aspect ratio before resizing to 224 × 224 pixels and ImageNet normalization for the ViT classifier. The classifier produces a four-class score vector and candidate label. Where segmentation is requested, a mask or overlay can be produced by the separate segmentation artefact.

The canonical detector labels are cattle, lumpy_skin, foot_and_mouth and mastitis. Class-order consistency must be verified against each checkpoint. A score is a model output, not verified clinical probability, because no calibration experiment is reported. The API design must also retain source image and crop identity so that a reviewer can determine which region produced the candidate class.

### 4.3.3 Core method

YOLOv8 provides architecture detector and training interface [18]. ViT represents images as patch sequences and transfers a pretrained visual representation to the classification task [19]. ResNet supplies the residual backbone used in the custom mask model [20]. These established architectures are applied to the project's dataset; the work does not claim to invent new backbone methods.

The detector dataset combines six Roboflow projects and harmonises their annotation labels. The reported training and validation annotation counts total 10,140 object instances: 6,350 cattle, 2,397 lumpy_skin, 946 foot_and_mouth and 447 mastitis. Instances are not unique-image counts. Class weighting, augmentation and oversampling address imbalance during the relevant training stages, although per-class improvements are not established by aggregate accuracy alone.

The detector has 111 recorded training epochs and a selected validation checkpoint at epoch 86. The ViT uses AdamW, weighted cross-entropy, mixed precision and early stopping. The mask workflow uses a pretrained backbone, initial freezing, later upper-layer fine-tuning and balancing. Figure 4.2 illustrates the research cascade and the optional nature of segmentation.

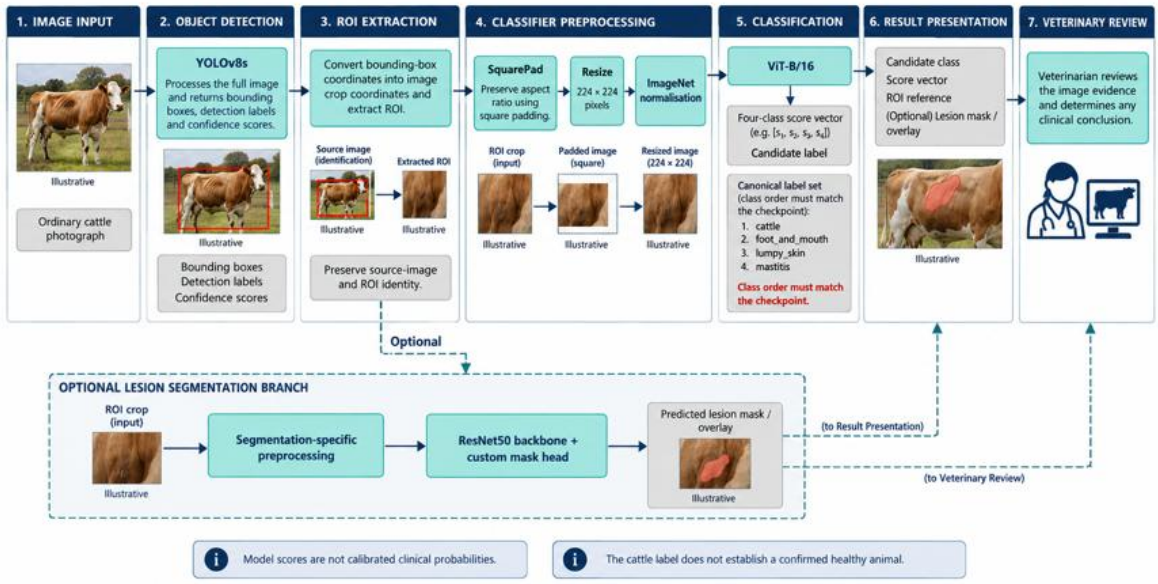



**Figure 4.2 Image preprocessing and inference workflow**

### 4.3.4 Contribution to the integrated solution

The component adds visual evidence to the farm-level record and supports veterinary review. It complements wellness triage, which identifies general physiological concern, and the spatial and forecasting components, which operate at event and

population levels. A proposed integration result should carry a case or analysis identifier, label mapping, model version, scores and image references. A clinician-reviewed record must remain separate from the initial prediction.

The trained stages and their metrics establish research feasibility. They do not establish that the final image cascade has been exposed through a tested ADRS production endpoint. The diagnosis-to-clinical-record handover is therefore an integration boundary requiring explicit verification rather than an already measured end-to-end result.

## 4.4 Component 2 Summary

### 4.4.1 Owner and subproblem

The Persistent Health Anomaly and Body Condition Scoring module was developed by W.M.P.J. Wijenayake (IT22194558). It addresses general wellness interpretation from productivity trends, body condition, static metadata and environmental stress. The component provides an automatic anomaly trigger and a manual observation trigger, allowing detailed triage to begin from either recorded deviation or farmer concern.

### 4.4.2 Inputs and outputs

Persistent monitoring uses daily milk yield, body weight, breed baselines and three-day trend features. Detailed triage uses seven days of eight physiological and environmental variables: ambient temperature, humidity, THI, body temperature, milk yield, water intake, feed intake and body weight. A 45-node encoded static vector represents relevant metadata, while a visual branch receives BCS derived from a two-dimensional rear-quarter image.

Outputs include anomaly feedback, BCS, a Healthy, Heat Stress or Unhealthy classification, a visual explanation and non-medical management guidance. The source interface occasionally uses broader At Risk terminology; the clinical class contract must preserve the three model labels and make any presentation mapping explicit. An Unhealthy result is not a confirmed infection or named pathology.

### 4.4.3 Core method

A Random Forest learns a seven-rule engineered anomaly architecture based on breed-relative and recent trend deviations. YOLOv8s localises the animal and a custom CNN estimates continuous BCS. The late-fusion network processes the temporal sequence through an LSTM branch and combines it with dense representations of static metadata and visual BCS. Multimodal learning provides a general framework for relating complementary inputs [21] ; the predictive benefit in this project must be assessed using its own experiments.

The dataset covers 40 breed baselines grouped into temperate, indigenous tropical and crossbred categories. Earlier days in some seven-day sequences are reconstructed by interpolation anchored to Day 7; the monitoring labels also depend on engineered scenarios. The experiments therefore assess behaviour within the constructed data process. They do not demonstrate prospective detection of naturally arising illness across all 40 breeds.

The explainability module adapts Grad-CAM using absolute gradients to generate anatomical heatmaps and a targeting crosshair. Grad-CAM is a gradient-based visual localisation technique [22]. The regression adaptation is a component contribution, but a plausible heatmap does not independently prove clinical correctness, causal reasoning or freedom from background bias. Figure 4.3 illustrates the dual triggers and the three fusion branches.

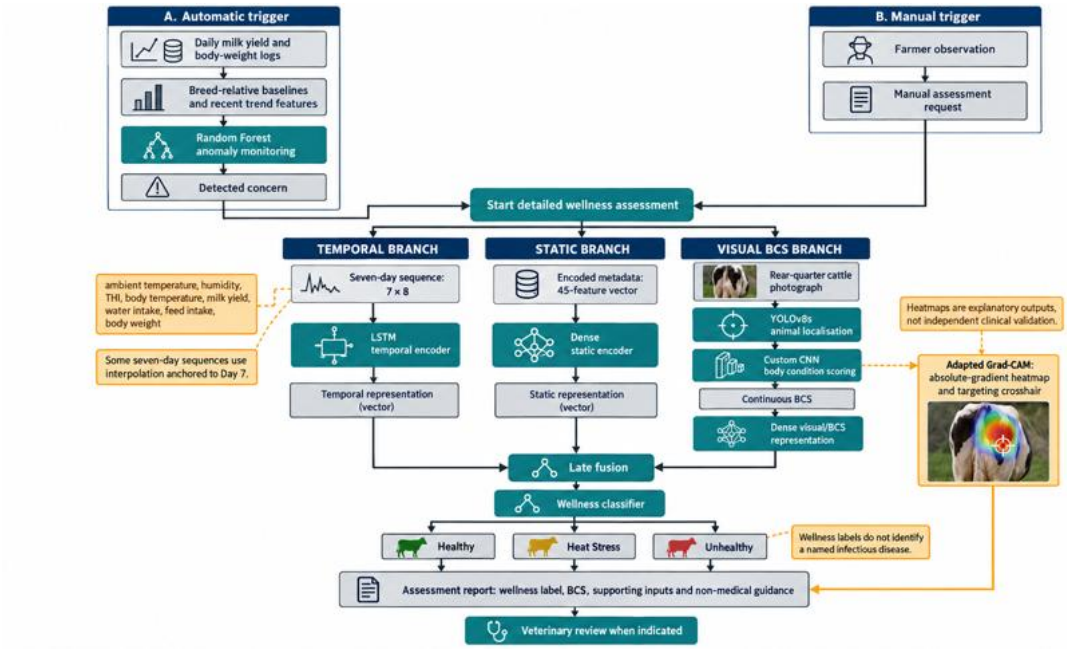



**Figure 4.3 Dual-trigger wellness assessment and multimodal fusion**

### 4.4.4 Contribution to the integrated solution

Wellness triage provides a structured first layer of animal observation. It can draw attention to sustained deviations, attach environmental context and guide further professional assessment. Its records can accompany image evidence and case review. The triage service does not itself create a confirmed geospatial disease event, and its general wellness labels must not be directly substituted for the FMD or LSD targets in district-month modelling.

The standalone BCS analyser also supports routine assessment outside the seven-day triage window. Its image-processing outputs help users understand the selected region. External weather dependence and synthetic sequence construction limit the evidence, so operational guidance must remain cautious and non-medical.

## 4.5 Component 3 Summary

### 4.5.1 Owner and subproblem

Real-Time Geospatial Clustering and Trajectory Mapping was developed by G.A. Sandaru (IT22258908). It translates geographically distributed livestock disease records into interpretable surveillance context. The component connects event identity, location, historical source context and veterinary service area, allowing a confirmed case to be examined alongside relevant spatial information.

### 4.5.2 Inputs and outputs

The research layer uses historical LSD and FMD records attributed to FAO EMPRES-i and WOAH WAHIS, retaining disease labels, dates, coordinates, provenance and data-quality indicators. Operational adapters read veterinarian-verified clinical cases and district context. Outputs include source markers, relative spatial cells, geometric direction descriptors, historical apparent-rate and nominal-reach information, and map panels.

National overview, My Area and Analysis and Trends interfaces offer different perspectives. Historical cell scores and Component 4 district-month probabilities are separate outputs with different target definitions, spatial supports and time horizons. The map should therefore label the selected disease, origin date and evidence type rather than placing all quantities under an undifferentiated risk label.

### 4.5.3 Core method

The research pipeline applies provenance-preserving normalisation, conservative duplicate treatment and controlled historical origin selection. ST-DBSCAN supports descriptive event grouping [15] . The evaluated LSD reference calculation constructs a geodesic area of interest and approximately 5 km grid cells, aggregating distance-decaying source contributions with a frozen 25 km exponential kernel.

The resulting surface ranks eligible cells relative to one another. A cell-local negative gradient supplies geometric direction context; a historical apparent spread-rate calculation supplies nominal reach. Environment-related adapters and alternative

candidates were studied, but environmental transport is not established by the distance-only reference score. These descriptors must not be interpreted as confirmed transmission routes or calibrated infection likelihood.

### 4.5.4 Contribution to the integrated solution

The component extends ADRS by providing a district-level forecasting perspective alongside animal-health monitoring, disease diagnosis and geospatial surveillance. Its interfaces support national and district-level risk assessment, forecast decision records, veterinary investigation follow-ups and advisory preparation. These functions allow forecasting information to contribute to veterinary decision-making without treating an estimated outbreak probability as confirmation of disease.

The research demonstrated stronger retrospective discrimination for FMD than for LSD under their respective evaluation procedures. The integrated prediction service provides probabilities, risk classifications, explanatory information and data-source indicators, while highlighting limitations associated with the available surveillance and environmental information.

*FIGURE 3.4 PLACEHOLDER*

*Insert screenshots of the national map and My Area view, with verified-case markers, selected historical origin, source inspection, relative cell score legend and direction/reach labels. Retain the analysis date and distinguish historical from operational data.*

**Figure 3.4: National and district-oriented geospatial surveillance views**

Source: Authors' own architecture or application evidence.

## 4.6 Component 4 Summary

### 4.6.1 Owner and subproblem

The Climate-Informed Livestock Disease Forecasting and District-Level Outbreak-Risk Nowcasting component was developed by S.S. Kumarasinghe (IT22221414). It addresses the need for district-level estimates of reported Foot-and-Mouth Disease (FMD) and Lumpy Skin Disease (LSD) outbreak occurrence by examining historical surveillance records alongside environmental, seasonal and spatial information. The component is intended to support veterinary surveillance and planning rather than diagnose disease in individual animals.

### 4.6.2 Inputs and outputs

The forecasting models use district-month surveillance information together with climatic variables such as rainfall, temperature, humidity and wind speed. Seasonal indicators, livestock density, large-scale climate indices and previous outbreak activity provide additional context. The documented data sources include CHIRPS for rainfall [23], ERA5 for atmospheric information [24], and DAPH epidemiological materials for disease surveillance [25]. The preparation and alignment of these sources were carried out as part of the component's dataset development.

The research datasets contain 2,400 FMD district-month observations covering 2017–2024 and 1,500 LSD observations covering 2020–2024. Previous-month outbreak indicators describe whether an outbreak was reported in the relevant district or neighbouring areas rather than the number of infected or deceased animals. The model outputs include estimated outbreak probabilities, LOW, MEDIUM or HIGH risk classifications, explanatory information and data provenance details. Additional Stage 2 information is provided where applicable.

### 4.6.3 Core method

The FMD prediction service uses Logistic Regression, with a 30-feature baseline model and an optional 31-feature variant that incorporates verified previous-month outbreak status for the same district. The LSD service uses a 28-feature Platt-calibrated

ElasticNet Logistic Regression model when the required outbreak-history information is available, with a 27-feature fallback model otherwise.

The models were evaluated using disease-specific temporal assessment procedures. FMD was assessed through expanding-window walk-forward evaluation, while LSD was evaluated using Leave-One-Year-Out validation across 2020–2024. As the LSD procedure can use observations from later years when evaluating an earlier year, its results should be interpreted as retrospective generalisation evidence rather than strictly forward-looking forecasting performance.

For prediction requests, the component uses an exact district-month feature record when available. Where suitable information is unavailable, historical same-calendar-month observations or district-level historical summaries may be substituted, with the source of the substituted information identified in the response. These mechanisms allow the service to produce assessments under incomplete data conditions, but they do not replace observed future weather information.

Model explanations are derived from the fitted linear models and their transformed input features. These explanations help users understand factors associated with a prediction but do not establish causal relationships. Additional uncertainty methods, including conformal prediction, were examined during the research rather than established as fully operational runtime features.

### 4.6.4 Contribution to the integrated solution

The component adds a district-level planning perspective to farm and event information. DAPH-oriented users can request national summaries, preserve forecast decision records and initiate investigation follow-ups; veterinary users can review assignments and draft advisories. This workflow demonstrates software use of the model without converting a district probability into a confirmed animal diagnosis.

The primary product result is an accessible prediction service with explicit variant and data-source handling. Its principal research result is stronger FMD discrimination than LSD under their respective evaluations. Figure 3.5 presents role-specific risk information with provenance and advisory status visible.

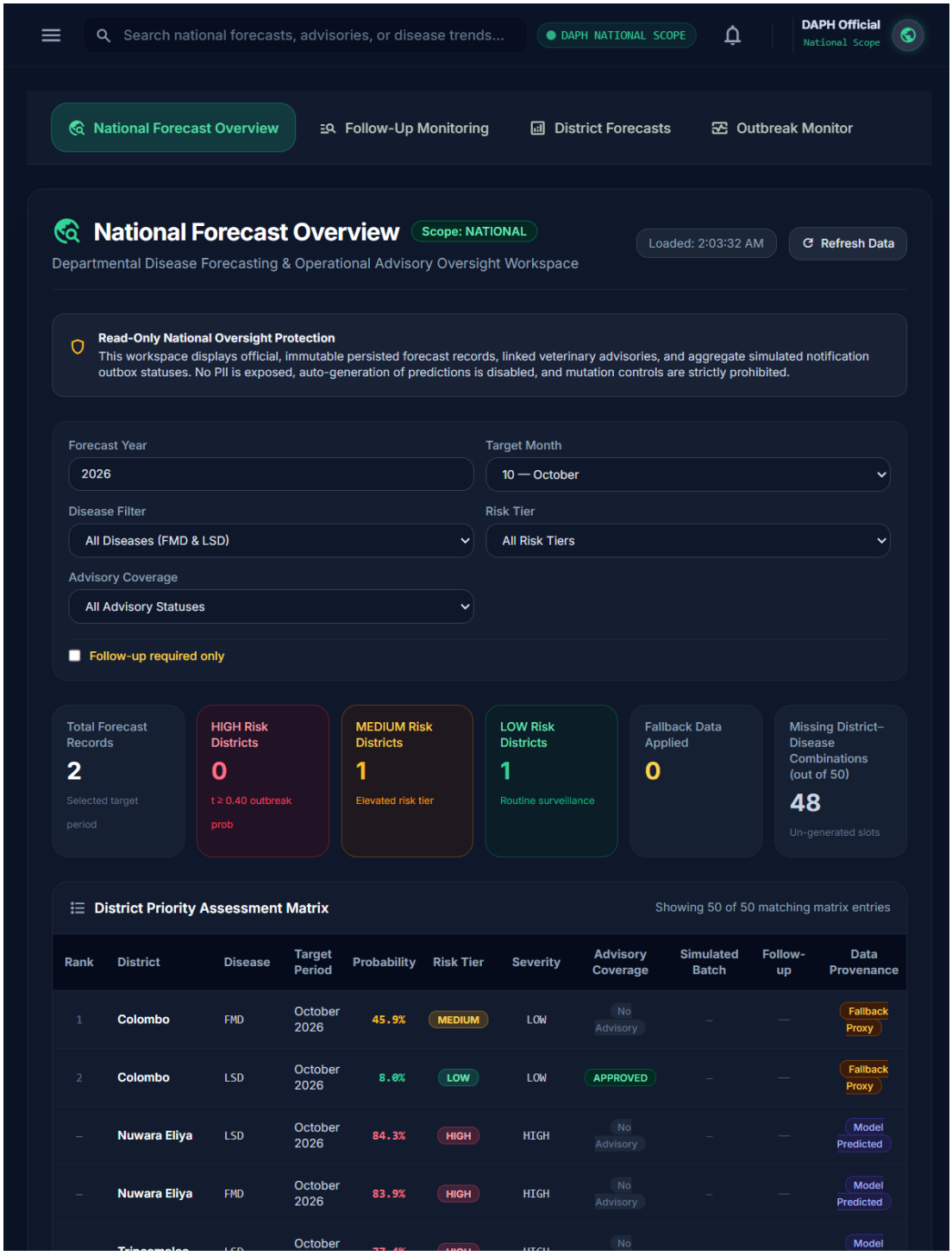



**Figure 4.4 National forecast overview -DAPH Officials**

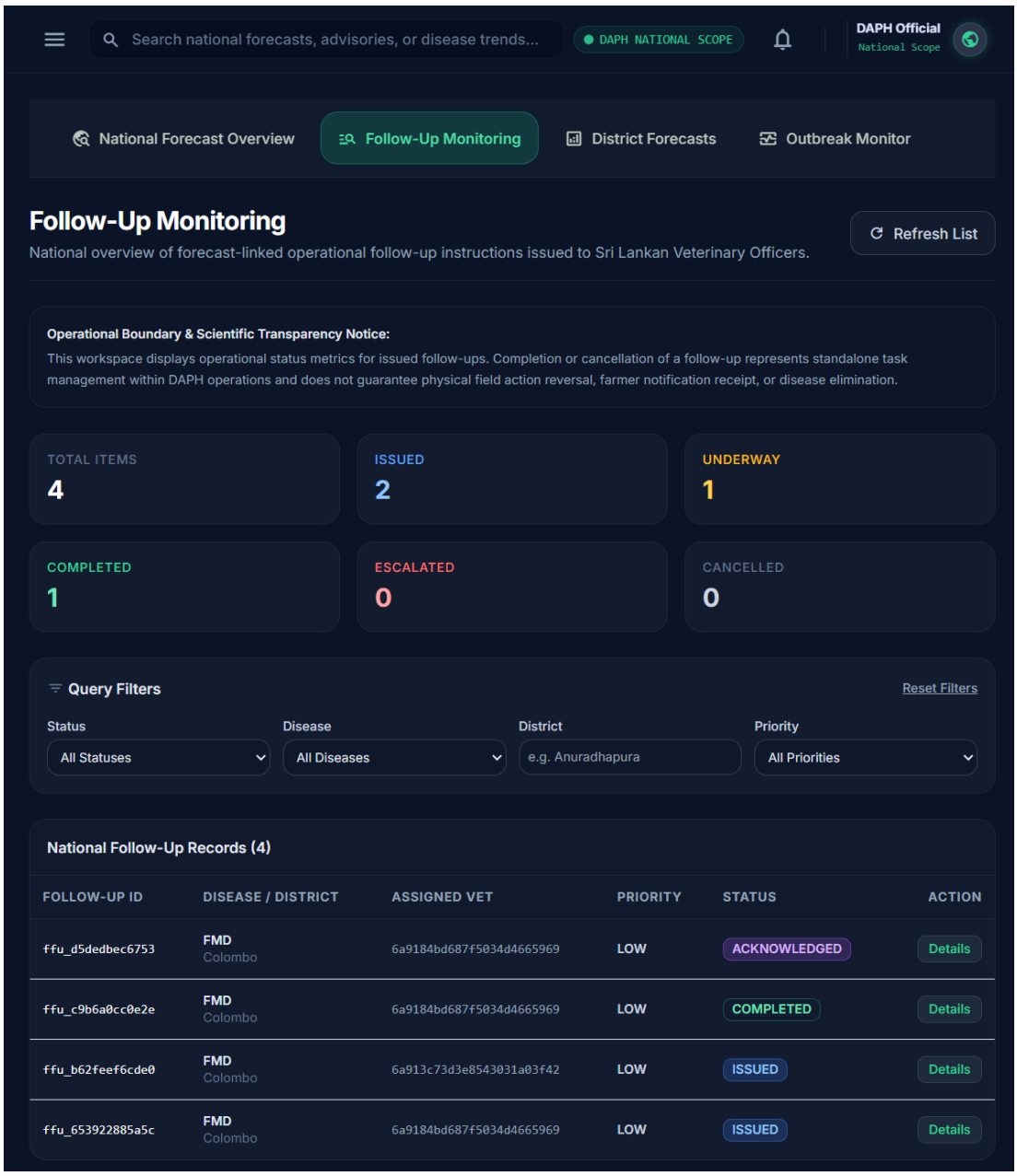



**Figure 4.5 :Follow up monitoring - DAPH officials**

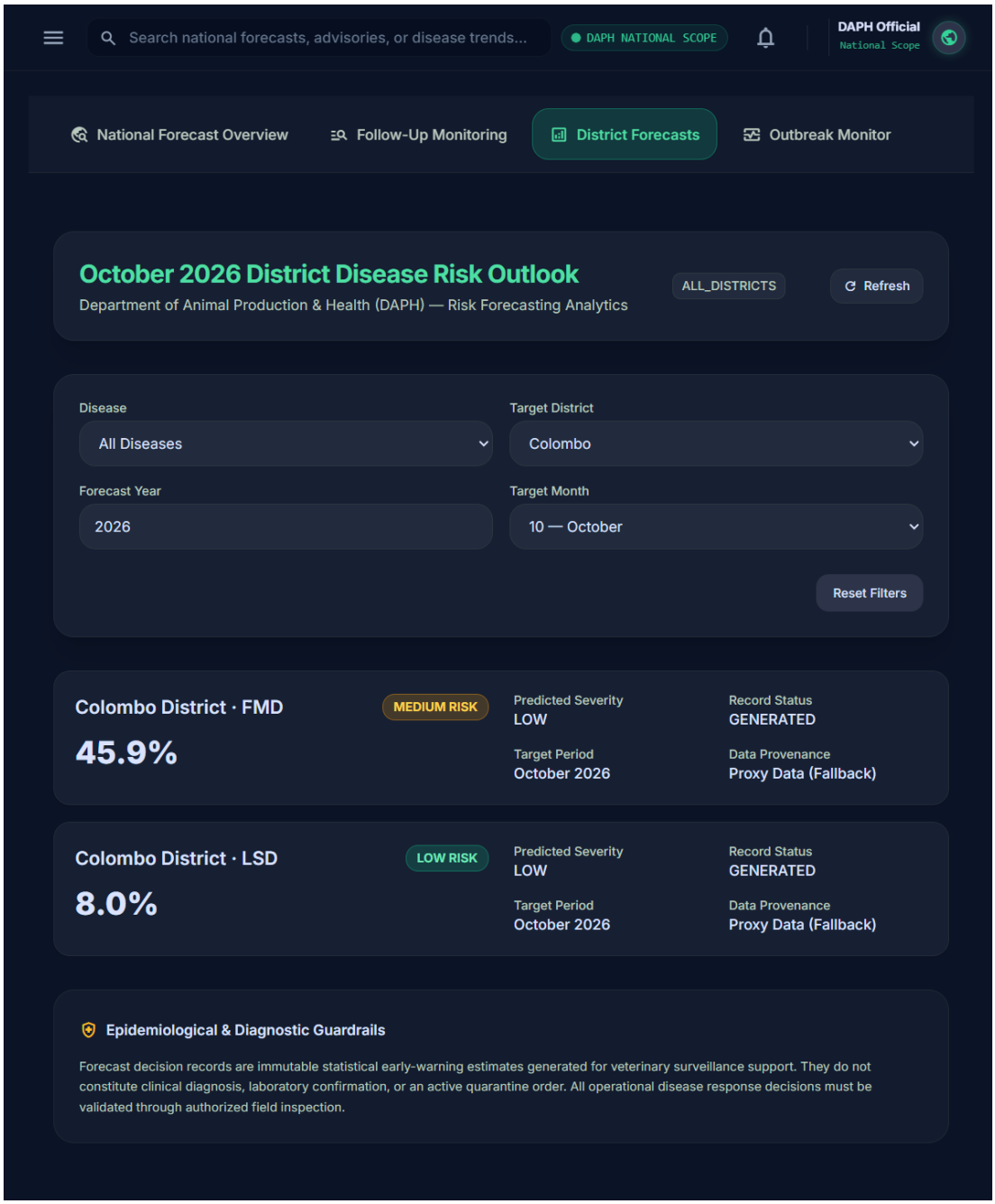



**Figure 4.6 : District forecast UI - DAPH officials**

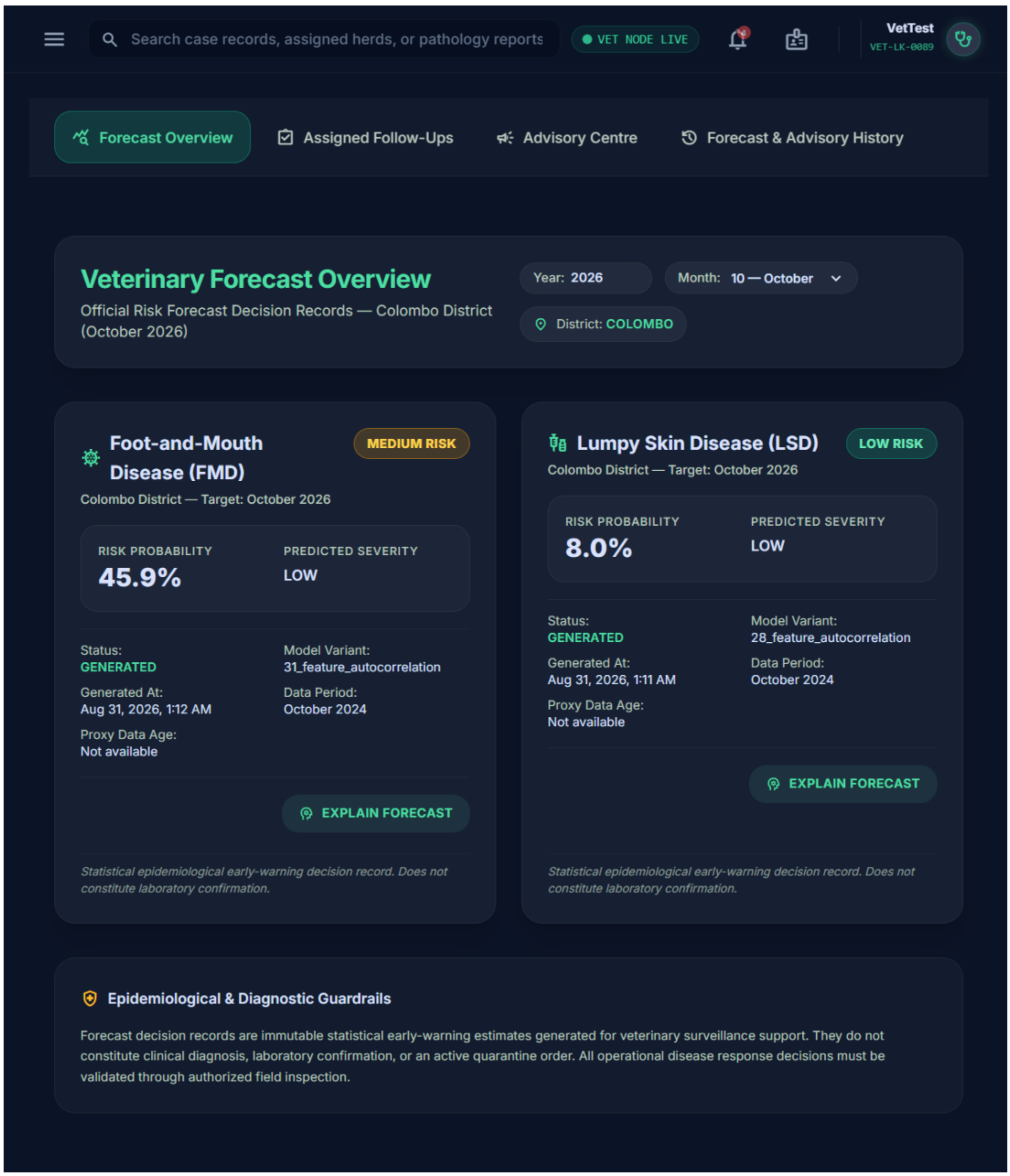



**Figure 4.7 : Veterinary Forecast Overview -Veterinary Dashboard**

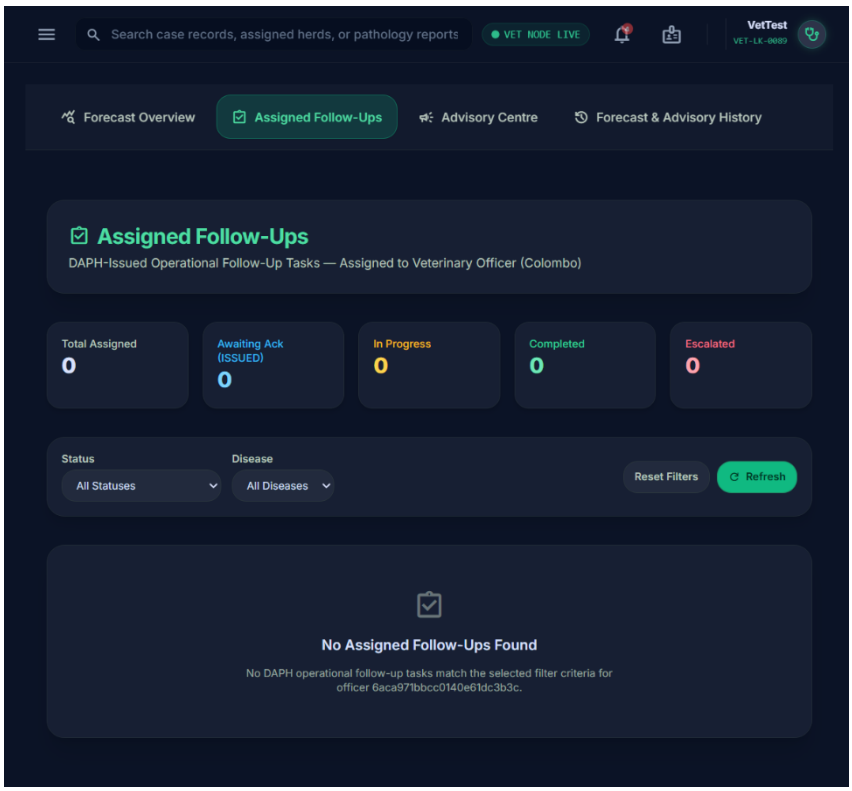



**Figure 4.8 : Assigned Follow ups - Veterinary Dashboard**

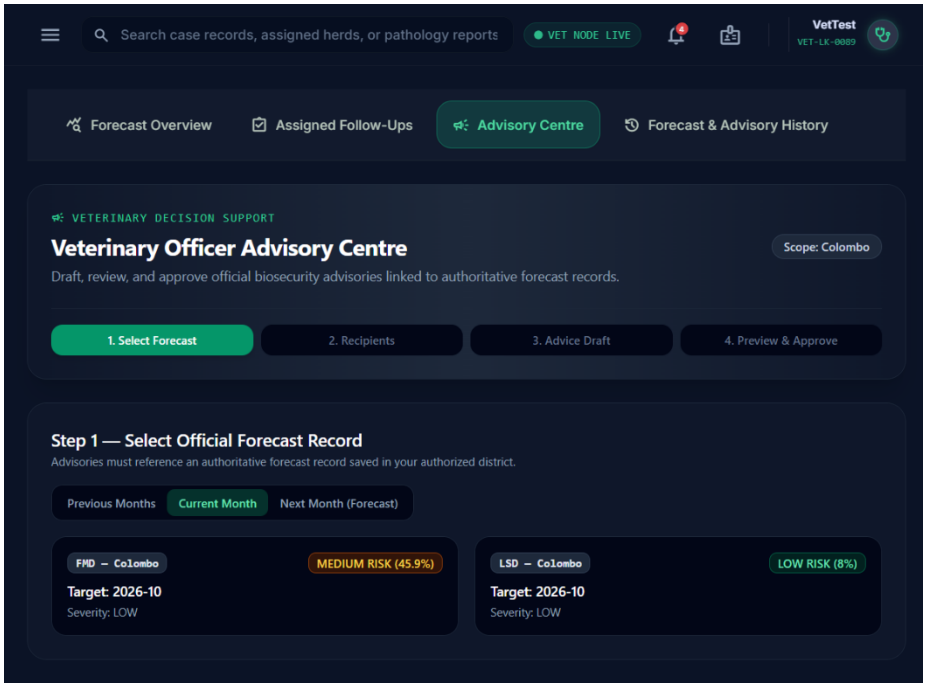



**Figure 4.9 Farmer Advisory Centre**

## 4.7 End-to-End User and Data Flow

The intended normal flow begins when a farmer records daily animal information. Persistent monitoring checks trends; a farmer observation or anomaly can initiate the seven-day triage workflow. The application gathers the required sequence, metadata, image-based BCS and weather context. The resulting wellness record is retained with its input period and analysis identity. A relevant disease photograph can also be analysed by the image pipeline and attached as candidate evidence.

A veterinary officer reviews the concern using clinical information and, where required, examination or laboratory investigation. Confirmation is an explicit professional action. A confirmed record includes disease, event date, animal or farm context and authorised location. The geospatial operational layer presents that verified event; it does not promote an unreviewed softmax result to a confirmed outbreak.

Spatial analysis can then provide historical source and relative-cell context. Separately, a surveillance user requests a disease, district, month and year from the forecasting service. The response identifies its model variant and whether exact-period or proxy features were used. Officials may preserve the assessment, create a follow-up and obtain an advisory draft. These actions remain distinct from prescribing treatment or enforcing disease-control measures.

The workflow is designed to allow professional review and correction at each transition. The recorded forecasting integration and geospatial views support parts of this flow; the entire triage-to-image-to-confirmation-to-surveillance chain has not been measured as a single clinical or performance experiment. Figure 4.4 depicts the intended sequence and identifies the outstanding image-service handover.

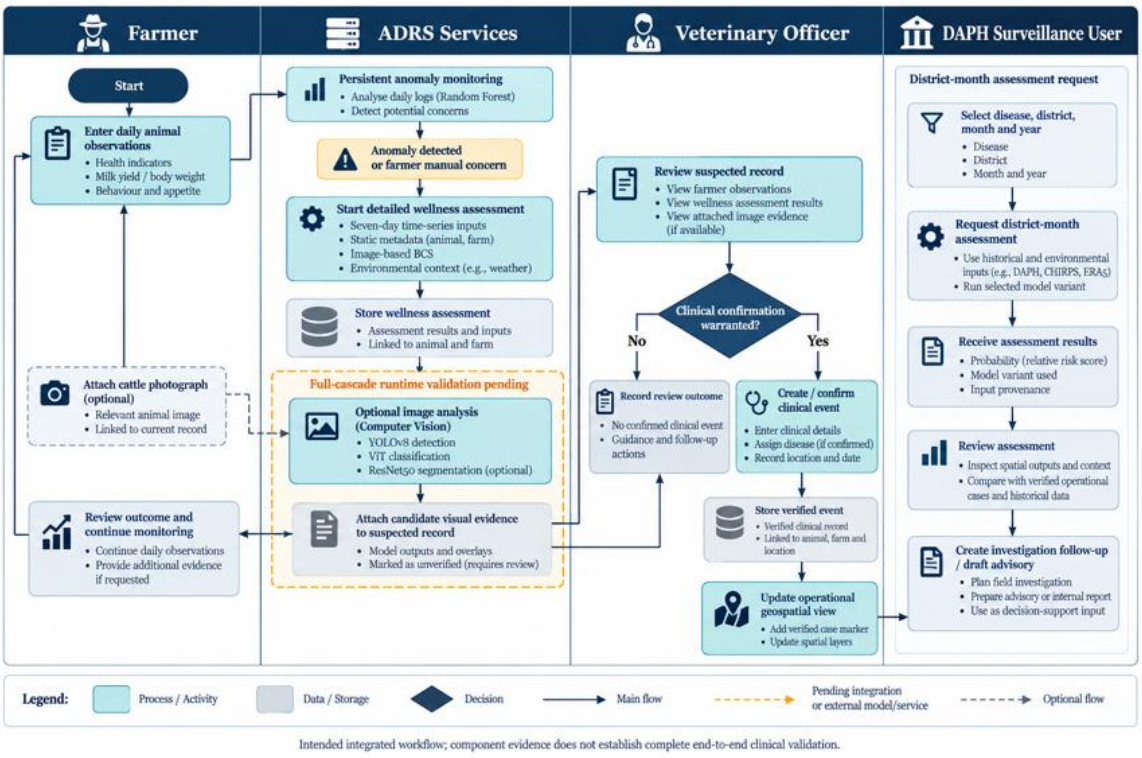



**Figure 4.10 End-to-end reporting review and surveillance workflow**

# 5. COMPONENT INTEGRATION AND COHESIVE OPERATION

## 5.1 Integration Strategy

The application follows modular integration within a shared repository and service framework. Each research component owns its processing pipeline, model or analysis artefacts, and domain-specific outputs. Shared services provide authentication, directory context, persistence and navigation. Component interfaces connect those responsibilities without requiring one research module to load another's internal training code.

The integration sequence consists of establishing common identities and roles, registering routes, validating request and response shapes, connecting views, preserving confirmed-case adapters, and exercising operational workflows. Frozen model weights, feature order, scalers and thresholds must remain separate from application integration changes. This separation prevents a route or interface adjustment from silently altering an evaluated scientific configuration.

The geospatial module separates historical scientific processing from operational verified-case access. The forecasting module separates FMD and LSD inference services from forecast records and follow-ups. The triage route aligns temporal and static inputs before inference. The smart-diagnosis pipeline has stage artefacts but still requires a demonstrated live service boundary. Table 5.1 records these different integration states rather than treating the existence of four components as a completed full-system test.

**Table 5.1 Component integration responsibilities and status**

| Boundary | Responsibility | Recorded status |
|---|---|---|
| Shared host services | Identity, roles, directory context and operational persistence | Application-level connections described |
| Triage service | Input assembly, scaling and wellness inference | Client/API flow and simulated batch reported |
| Diagnosis service | Detector, classifier and mask orchestration | Stage artefacts available; full runtime handover pending |
| Geospatial adapters | Verified cases, historical analyses and district views | Operational views and interface contracts described |
| Forecasting services | Frozen prediction and cross-role records or follow-ups | Live HTTP workflow and recorded tests reported |

## 5.2 Interfaces, Data Contracts and Shared Models

The shared contract must preserve animal ID, farm ID, user ID, case ID, district identity and relevant observation dates. Historical source-event identifiers must remain separate from operational case identifiers. Disease names require a canonical mapping, since detector labels, clinical disease names and forecasting targets have different schemas. Appendix A gives a consolidated interface specification and validation rules.

Triage uses /api/monitor/predict-7day for the detailed inference workflow. The temporal payload is aligned to a batch tensor of $1 \times 7 \times 8$, the static metadata to $1 \times 45$, and BCS to the visual input expected by the model. Feature order and units are part of the contract. Scaling examples in the component implementation include temperature divided by 50.0 °C and milk yield divided by 50.0 L; these are recorded model preprocessing choices, not universal veterinary constants.

For diagnosis, the minimum service output should include image and crop identity, bounding boxes, canonical class names, candidate scores, preprocessing identity, model versions and an optional mask reference. The final trained class-index order must be verified before the API can be treated as stable. No deployed endpoint path is asserted where the component evidence does not supply one.

Geospatial HTTP routes are relative to /api/geospatial. Table 5.2 lists their principal functions. GeoJSON uses WGS84 coordinates in [longitude, latitude] order, consistent with RFC 7946 [26]. A coordinate-order check is necessary because an otherwise valid numeric pair can place a marker in the wrong location.

**Table 5.2 Principal geospatial interfaces**

| Interface | Purpose and boundary |
|---|---|
| GET /origins and /origins/{id}/trigger-sources | Eligible historical origins and source geometry |
| GET /analysis/{id}/summary, /cells and /sources | Historical LSD analysis and source evidence |
| GET /analysis/{id}/fmd-risk | Disease-specific historical FMD spatial score |
| WebSocket /ws | Coherent versioned retrospective snapshots |
| GET /operational-context | Authorised veterinarian-verified case context |
| GET /operational-events/stream | Verified-case operational event stream |
| GET /my-area | Veterinary district-oriented context |
| GET /analysis-trends | Retrospective research aggregates |

Forecasting paths are relative to /api/v1/risk-forecasting. Table 5.3 identifies the main endpoints. Pydantic validates district, year, month and applicable variant selections. Prediction responses include probability, labelled category, model-related explanations, provenance and conditional Stage 2 information. The outbreak-status interface permits occurrence information to be shared without coupling another component directly to model internals.

**Table 5.3 Principal forecasting and operational interfaces**

| Interface | Purpose |
|---|---|
| POST /predict/fmd and /predict/lsd | Individual district-month probability estimation |
| POST /forecast/fmd and /forecast/lsd | National district-wise climatological risk projections |
| GET /outbreak-status/{disease}/{district}/{year}/{month} | Occurrence-status interface for component exchange |
| POST /records | Preserve a forecast decision record |
| POST /follow-ups | Create a veterinary investigation follow-up |
| POST /advisories | Create a farmer advisory draft |

Input validation must distinguish missing values from genuine zeroes and reject unsupported diseases, invalid periods and incompatible tensor shapes. Errors require a useful status and correlation identifier; the response must not substitute a fabricated safe result. Snapshot origin and version checks prevent mixed spatial responses, while model-feature contracts prevent reordered forecasting or triage inputs.

## 5.3 Shared Nonfunctional Requirements

The common nonfunctional requirements concern performance, usability, security, privacy, reliability, maintainability, scalability and accessibility. Table 5.4 summarises

their implementation evidence and remaining verification. These requirements apply to the complete platform; an isolated model latency or one test suite cannot establish compliance for every component.

**Table 5.4 Shared nonfunctional requirements and verification needs**

| Requirement | Design or recorded support | Remaining verification |
|---|---|---|
| Performance | Async request/database access and cached model loading | Complete-workflow latency, percentiles and concurrent load |
| Usability | Responsive role-oriented views and labelled outputs | Representative farmer and veterinary task evaluation |
| Security | JWT, role checks and validation in selected services | Consistent endpoint enforcement and adversarial access tests |
| Privacy | Role/district scoping and aggregated forecasting | Farm-location protection, retention and access audit |
| Reliability | Validation, explicit fallbacks and unavailable states | Dependency faults, recovery and full-chain consistency |
| Maintainability | Component routes, services and model artefacts | Versioned contracts, dependency locks and release reproducibility |
| Scalability | Separate workloads and async database access | Inference scheduling, worker memory and horizontal-load tests |
| Accessibility | Text labels and responsive presentation | Keyboard, contrast and assistive-technology assessment |

The triage source sets a 10–15 second full-workflow response target and 99.5% availability target. These are design targets rather than demonstrated platform outcomes. Its 186.4 ms reported batch inference timing concerns a narrower operation. The image pipeline has no complete service-latency record; geospatial streaming and district-wise prediction also require workload-specific performance assessment.

Security claims must be proportional to the available evidence. The presence of JWT or a protected route does not establish enforcement across every API. Detailed farm coordinates, contact information and clinical records require backend authorisation independent of frontend visibility. The intended privacy framework must consider Sri Lanka's Personal Data Protection Act and its 2025 amendment [27]; the research does not claim a completed statutory compliance audit.

WCAG 2.2 provides an appropriate accessibility reference for later validation [28]. Readable labels, visible focus, keyboard operation, non-colour risk cues and usable error messages should be assessed on actual tasks. Formal conformance cannot be inferred merely from responsive CSS or a dark theme.

### 5.4 Integrated Workflow and Exception Handling

Normal operation preserves a clear sequence from observation to analysis, review and confirmation. Missing daily logs, invalid images, unavailable weather, failed model loading, unknown spatial origins and incomplete surveillance features require different responses. The integration design therefore uses explicit validation and provenance rather than a common catch-all prediction.

For triage, zero historical yields require guarded percentage calculations. The documented guard avoids division by zero by returning a zero drop in that calculation, but this can conceal missing or unsuitable baseline information. A deployment should retain a separate missing-baseline indicator and request review rather than equating computational safety with animal health. An incomplete seven-day series must not be silently treated as seven measured days.

For image analysis, failure to detect a suitable ROI should lead to recapture or professional review. A poor-quality or unfamiliar image must not be reported as

healthy because no familiar disease class is selected. Since OOD rejection is not validated, the interface must not promise reliable automated rejection of every unrelated image. Mask generation remains optional; its failure should leave the classifier evidence identifiable and disclose that segmentation was unavailable.

For geospatial analysis, empty case sets produce an empty state rather than an invented location. Invalid origin identifiers and unavailable spatial analysis produce explicit statuses. Chunked snapshots must retain origin and version identity so that a delayed response from an earlier request does not overwrite a newer selection. Historical analysis remains visibly distinct from the current case layer.

For forecasting, an unavailable same-district lag triggers the supported fallback variant. A missing exact-period climate row may trigger a historical proxy with the substitution identified. If essential artefacts cannot load, prediction should be unavailable. Conditional Stage 2 information is advisory; LSD Stage 2 is mainly a quiet-period suppressor and has not established reliable active-outbreak severity discrimination.

Cross-component recovery requires auditability. Case verification and forecast decision records should retain actor, time and record identity. Retries must not duplicate a confirmed case or investigation. Browser disconnection must not convert a partial response into a completed scientific snapshot. These behaviours define acceptance tests in Chapter 6 and Appendix C; they are requirements where complete implementation evidence is not yet established.

## 5.5 Team Contribution and Dependency Matrix

Table 5.5 summarises individual and shared responsibilities. The contributions connect research artefacts to the common product but do not imply that every researcher independently validated all other components. Appendix B records the team contribution declaration and sign-off fields.

**Table 5.5 Team contribution and dependencies**

| Owner | Main research and development | Main dependency or shared handover |
|---|---|---|
| A.L.M. Athulathmudali | Dataset harmonisation; YOLOv8s, ViT and mask experiments; crop/inference utilities | Class mapping and image evidence for veterinary review |
| W.M.P.J. Wijenayake | Breed/trend features; Random Forest; BCS CNN; LSTM late fusion; Grad-CAM and triage views | Animal logs, weather context and wellness record |
| G.A. Sandaru | Source provenance; spatial ranking and descriptors; geospatial APIs and map views | Verified clinical cases and veterinarian district context |
| S.S. Kumarasinghe | District-month reconstruction; FMD/LSD models; temporal evaluation; prediction and follow-up views | Surveillance/climate files, frozen features and shared role services |
| Team collectively | Common navigation, identities, persistence boundaries and reporting | Release-wide regression, confirmation workflow and field evaluation |

# 6. INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

## 6.1 Common Technology Stack and Infrastructure

ADRS uses a web-based architecture with React, Vite and Tailwind CSS on the client, Python and FastAPI on the backend, and MongoDB for operational persistence. Motor and PyMongo provide database access. The common stack allows component routes and interfaces to be organised within one application while retaining separate scientific processing and artefacts. Table 6.1 distinguishes common services from the research-specific libraries.

**Table 6.1 Common and component-specific technologies**

| Layer or component | Technologies | Principal purpose |
|---|---|---|
| Presentation | React, Vite, Tailwind CSS | Responsive views, shared navigation and workflow state |
| Application services | Python, FastAPI, Uvicorn, Pydantic | Routing, validation and service orchestration |
| Operational persistence | MongoDB, Motor, PyMongo | User, farm, case and follow-up records |
| Wellness | Scikit-learn, TensorFlow/Keras, YOLOv8, OpenCV | Monitoring, BCS, late fusion and explanation |
| Smart diagnosis | Ultralytics, PyTorch/torchvision, TensorFlow/Keras | Detection, ViT classification and custom mask model |
| Geospatial | MapLibre GL, Leaflet and geospatial utilities | Geometry processing, spatial results and map presentation |

| | | |
|---|---|---|
| Forecasting | Scikit-learn, Pandas, NumPy | Tabular preparation, probability models and explanations |
| Verification | Pytest, httpx, Vitest | Selected backend, route and frontend checks |
| Collaboration | Git and GitHub | Component branches and integration history |

The triage implementation reports Python 3.10.12, TensorFlow/Keras 2.13, PyTorch 2.0, Scikit-learn 1.3 and a React 18 frontend with Vite 5 and Tailwind CSS 3.4. These are component-reported development configurations, not a verified dependency lock for the final whole application. Other artefacts require their own compatible environment, particularly the smart-diagnosis checkpoint and segmentation pipeline.

The smart-diagnosis thesis records an NVIDIA GeForce RTX 3050 Laptop GPU and approximately 203 minutes 52 seconds of ViT training. That training duration describes development hardware and is not a deployed request-time benchmark. The segmentation implementation is a custom ResNet50 mask head, rather than full Mask R-CNN, and uses TensorFlow/Keras despite the classifier's PyTorch implementation.

MongoDB is the operational store consistently described in the integrated implementation and triage implementation chapter. The triage design also discusses Azure Cosmos DB, and references both OpenWeatherMap and Open-Meteo weather access in different places. These are configuration and source-contract issues: the final release must identify its actual database deployment and active weather provider. The common report does not assume that both alternative database descriptions represent completed simultaneous infrastructure.

## 6.2 Development and Integration Process

Development proceeded through component-specific research and software work in a shared frontend/backend project. The group development account describes separate feature and integration branches and five pull-request merge commits between May and August 2026. This supports a history of progressive integration, but commit counts do not establish review quality, a complete Scrum process or successful release-wide testing.

The research process follows the design-science pattern of identifying a problem, defining objectives, developing an artefact, demonstrating it, evaluating it and communicating the results [29]. Each component applies that pattern to its own evidence: constructed wellness data, image-model experiments, retrospective spatial processing and district-month classification. The shared engineering process then connects those artefacts to roles, records and application services.

The triage implementation derives three-day drop and breed-baseline features, applies zero-division guards and serialises Random Forest and Keras artefacts. Model caching and loading with compile=False reduce unnecessary reinitialisation in the documented design. The React workflow coordinates wellness intake, seven-day scan and report views, while a typed-array decoder converts image Data URLs to binary payloads. Exact deployment-wide speedups are not inferred from those design choices.

Smart-diagnosis implementation harmonises six image-source projects, remaps classes, extracts annotation-guided crops and trains the three stages separately. Square padding avoids stretching a rectangular ROI to the ViT input aspect ratio. The saved detector and classifier checkpoints are best.pt and best_vit_model.pth. The separate detector and classifier class orders, hard-coded paths and missing live orchestration require resolution before the stage artefacts can become a reliable shared service.

Geospatial integration registers routes and connects the React feature to host adapters for verified cases, veterinary identity and district context. Its research outputs retain historical source identity and snapshot version. Forecasting integration registers dedicated services and schemas, connects role-oriented views, and supports forecast

decision records, follow-ups and advisory drafts. Model files, feature order, scaling and operational thresholds remain frozen at the recorded scientific configuration.

The forecasting thesis reports successful live HTTP cross-role execution and MongoDB-backed authentication at an integration checkpoint. This is stronger evidence than the existence of route code alone. Nevertheless, browser acceptance and additional smoke validation remained open at that checkpoint. The component result therefore supports its recorded workflow, not an automatic claim that the complete four-component system is clinically or operationally finished.

### 6.3 Deployment and Operational Configuration

The integrated development configuration requires frontend dependencies, backend dependencies, MongoDB connection settings, authentication configuration, model paths and component data artefacts. FastAPI is launched through Uvicorn and the client is built or served through Vite. Configuration must supply compatible model/scaler pairs and feature schemas; a database connection alone is insufficient to establish service readiness.

The triage thesis describes Microsoft Azure deployment, Docker, GitHub Actions, Azure Container Registry, health checks and managed hosting. The broader group account does not establish a completed production deployment of every component. These statements can be reconciled by treating Azure/container deployment as component-reported infrastructure and keeping whole-platform production readiness separate. No platform-wide uptime, backup-restoration or drift-monitoring result is claimed.

For a reproducible release, the required operational configuration consists of a pinned application commit, dependency locks, environment-variable documentation, model and data checksums, feature-order definitions and startup readiness checks. Authentication keys and connection strings must remain outside source control. Each service must expose a useful unavailable status when its own model or input files are missing, without allowing an otherwise healthy database check to conceal that failure.

The deployment plan should separate expensive image workloads from lightweight tabular prediction where necessary, while maintaining record identity across requests. Model caching must be considered per worker because multiple application processes can multiply memory usage. Request limits, inference timeouts, file validation and restricted cross-origin configuration require verification against the release selected for deployment.

Operational monitoring should record request failures, model loading errors, latency, unavailable weather or surveillance data, proxy substitutions and veterinary correction of model suggestions. Backup procedures should cover operational records, model artefacts and prepared-data lineage, with a tested restore in a separate environment. Notification delivery must also remain explicit: the forecasting component currently describes a simulated provider, so an advisory draft is not evidence of a message delivered to a farmer.

### 6.4 Evidence of Team Execution

Team execution is evidenced by individual artefacts, interface descriptions, reported branch integration and component-level testing. The diagnosis contribution includes dataset remapping, model checkpoints and evaluation histories. The triage contribution includes engineered datasets, model files, intake/report views and the 20-profile verification account. The geospatial contribution includes source-selection logic, map services and clinical adapters. The forecasting contribution includes district-month files, model variants, prediction contracts and operational workflow tests.

The division of responsibility is documented in Table 4.5 and Appendix B. Shared work includes navigation, user and veterinary context, persistence conventions and the reporting framework. The retained evidence supports component contributions and integration history; it does not establish a complete set of sprint and review records. A contribution declaration and signed academic certification require the authors' own review and signatures.

The final technical handover should tie each result to the exact artefact that produced it. Component checkpoint tests, integration branch tests and later release tests are

different execution states. Preserving their commit, environment and date avoids treating a historical pass count as a universal guarantee. Appendix C summarises the test evidence and the additional acceptance tests needed for the complete workflow.

# 7. INTEGRATED TESTING, EVALUATION AND VALIDATION

## 7.1 Evaluation Framework and Success Criteria

Evaluation is organised at three levels: component research, application integration and field effectiveness. Component research asks whether the model or analysis behaves usefully under its recorded data and protocol. Application integration asks whether records, roles and outputs move through defined interfaces correctly. Field effectiveness asks whether the complete workflow improves decision quality or reporting practice under real conditions. The current evidence is strongest at the first level and selected parts of the second.

Table 7.1 presents the component data and protocols. Results from different tasks are not combined into a single ADRS accuracy. The wellness classifier, image classifier, spatial ranking and district occurrence model have different labels, denominators and sampling processes. Internal project metrics are findings of the respective experiments, rather than claims attributed to unrelated external publications.

**Table 7.1 Evaluation data and protocols**

| Evaluation area | Data and split | Main measures and limitations |
|---|---|---|
| Persistent monitoring | Engineered breed/trend data; reported 80/20 split | Accuracy; independence from label rules needs assessment |
| Late-fusion triage | Constructed seven-day sequences; reported 80/10/10 split; 20 simulated profiles | Class agreement and narrow inference timing; not a clinical cohort |
| BCS regression | Two-dimensional images and expert-score comparison proposed | MAE/RMSE required; numerical results not established |

| | | |
|---|---|---|
| Detection/classification | Multi-source annotations and ROI crops; separate stage evaluation | Approximate YOLO validation metrics and ViT test accuracy |
| Segmentation | Baseline and refined evaluation on 100 samples | mIoU; split lineage and repeated runs need confirmation |
| Geospatial analysis | Historical sources and origin-based retrospective calculation | Relative ranking, geometric descriptors and interface checks |
| FMD risk | 2,400 district-months; walk-forward 2022–2024 folds | ROC-AUC, PR-AUC and held-out 2024 threshold results |
| LSD risk | 1,500 district-months; five LOYO folds, 2020–2024 | Mean ranking measures; not forward-only forecasting |

The group success criteria are the documented achievement of GO1–GO4 within their experimental scope, a consistent review-to-confirmation data boundary under GO5, and system-level verification under GO6. A target met in a research split is recorded as such. In particular, the 85% image classification target is met on aggregate held-out accuracy, but external-farm performance remains unevaluated.

For rare outbreaks, ROC-AUC and PR-AUC provide complementary views. Precision measures the fraction of alerts corresponding to observed positives; recall measures the fraction of observed positives identified. A lower threshold can increase alerts and outbreak recall simultaneously. Precision-recall analysis is particularly relevant to imbalanced outcomes [16]. Evaluation against later periods and location-aware splits is necessary where observations are dependent [17].

## 7.2 Integration and End-to-End Testing

The forecasting component reports 862 passing automated tests, comprising 371 backend and 491 frontend tests, with no failures in the recorded execution. The account includes prediction behavior, authentication, route handling, persistence and frontend functionality. It also describes successful live HTTP forecast-record and cross-role workflow execution. These are checkpoint-specific results and require rerunning against the final release before they can establish its status.

The triage component reports Pytest checks for scaling, dimension guards and zero-baseline handling, httpx route/database checks and Vitest frontend checks. Its 20-profile verification reports successful inference for every profile. The geospatial account identifies tests covering canonicalisation, source eligibility, spatial cells, direction identities, HTTP/WebSocket snapshots, role-scoped context and map interaction. It does not provide one release-wide total that can be added to the forecasting count without overlap.

The diagnosis component provides trained checkpoints, crop/inference utilities and stage evaluation. It does not supply a verified live ADRS API, production latency, security test or full-system handover result. A complete test must pass a known image through detection, actual predicted cropping, label mapping, classification, optional mask generation, record creation and veterinary confirmation. Stage test accuracy based on annotation-guided crops cannot replace this experiment.

Table 7.2 synthesises the recorded status of major test areas. Appendix C provides the acceptance cases needed to close remaining gaps. A proposed expected behaviour is identified as a requirement rather than recorded as a successful result.

**Table 7.2 Integrated testing evidence and open acceptance areas**

| Test area | Recorded evidence | Group-level conclusion |
|---|---|---|
| Forecasting and shared-role checkpoint | 371 backend plus 491 frontend passes; live HTTP workflow | Recorded pass at that checkpoint |
| Triage input and inference | Shape/scaling guards and 20-profile simulated run | Component behaviour reported |
| Geospatial contracts | Processing, snapshot and role-view tests described | Component tests described; final suite total unavailable |
| Diagnosis full cascade | Stage artefacts and separate metrics | Runtime end-to-end result pending |
| Confirmation-to-map workflow | Verified-case adapters and operational views | Interface support described; full-chain regression needed |
| Security across all endpoints | Selected authentication and role tests | Consistent release-wide enforcement not established |
| Performance and recovery | Narrow triage timing; explicit fallbacks | Full-workflow load and fault tests pending |
| User acceptance | Role-oriented interfaces; qualitative triage feedback | Structured whole-system field study pending |

Security testing should include cross-user farm-record access, cross-district veterinary access, expired or missing tokens and direct API access to hidden frontend functions. Reliability testing should include weather timeout, missing checkpoint, incomplete sequence, stale spatial snapshot, database interruption and duplicate confirmation retry. The research account does not supply a complete penetration or fault-injection report for the platform.

Usability evaluation should observe representative tasks: enter a daily log, interpret a triage result, attach a clinical photograph, review a suspected case, confirm an event and interpret a historical-proxy district estimate. Completion rate, errors, review time, escalation decisions and user comprehension are more informative for the integrated workflow than screenshot appearance alone. Human browser acceptance and professional review remain distinct from automated frontend tests.

## 7.3 Overall Results

## 7.4 Wellness and body condition results

The persistent Random Forest monitoring experiment reports 99.80% test accuracy on mathematically engineered breed-baseline and trend data. This demonstrates reproduction of the constructed label process under the recorded split. Because scenarios and rules shape both training data and targets, the result must not be interpreted as 99.80% sensitivity to naturally arising disease. A prospective longitudinal study is necessary to measure false alerts and lead time before clinical concern.

The 20-profile comparison reports 100.0% classification agreement for the approximately 40,000-parameter base late-fusion model and 70.0% for the approximately 209,000-parameter expanded model. The component attributes the difference to the expanded model's difficulty with noisy thermal-stress scenarios. These values describe 20 simulated profiles, not a representative population or a statistically established general advantage of small networks. A model-selection batch should also be followed by an independent final test.

The same protocol reports 186.4 ms mean late-fusion inference latency and 20/20 successful inferences. The reported 0.00 MB memory change over the batch is a bounded observation at the measurement resolution; it does not prove leak-free long-running service operation. Numerical BCS MAE or RMSE is not established, and heatmap plausibility does not substitute for a regression benchmark or blinded expert evaluation. Table 7.3 preserves these evidence limits.

**Table 7.3 Wellness component results and interpretation**

| Experiment | Reported result | Interpretation |
|---|---|---|
| Random Forest monitoring | 99.80% accuracy | Engineered-data test; clinical performance unknown |
| Base late-fusion model | 100.0% on 20 simulated profiles | Small constructed verification batch |
| Expanded late-fusion model | 70.0% on the same 20 profiles | Comparative behaviour within that batch |
| Late-fusion timing | Mean 186.4 ms; 20/20 successful | Narrow inference operation, not full platform latency |
| Memory observation | 0.00 MB reported change | Limited observation; long-run stability unproven |
| BCS and Grad-CAM | Score/heatmap outputs illustrated | Numeric regression error and explanation validity pending |

Figure 6.1 presents the monitoring and 20-profile comparison evidence, identifying the synthetic evaluation context. These results describe experimental behaviour rather than clinical effectiveness.

*FIGURE 6.1 PLACEHOLDER*

*Insert the Random Forest evaluation output and the base-versus-expanded comparison from the triage thesis. Retain 99.80%, 100.0% and 70.0%, and label the 20-profile comparison as simulated.*

**Figure 6.1: Engineered-data monitoring and simulated triage comparison**

## 7.5 Image analysis results

The updated diagnosis experiment reports YOLOv8s validation precision and recall of approximately 0.80, mAP@50 of approximately 0.81 and mAP@50–95 of approximately 0.54 around the best checkpoint at epoch 86. These values are read from training curves and remain approximate. Exact exported results and per-class scores are required before presenting additional decimal precision. Figure 7.1 presents the original training and validation curves.

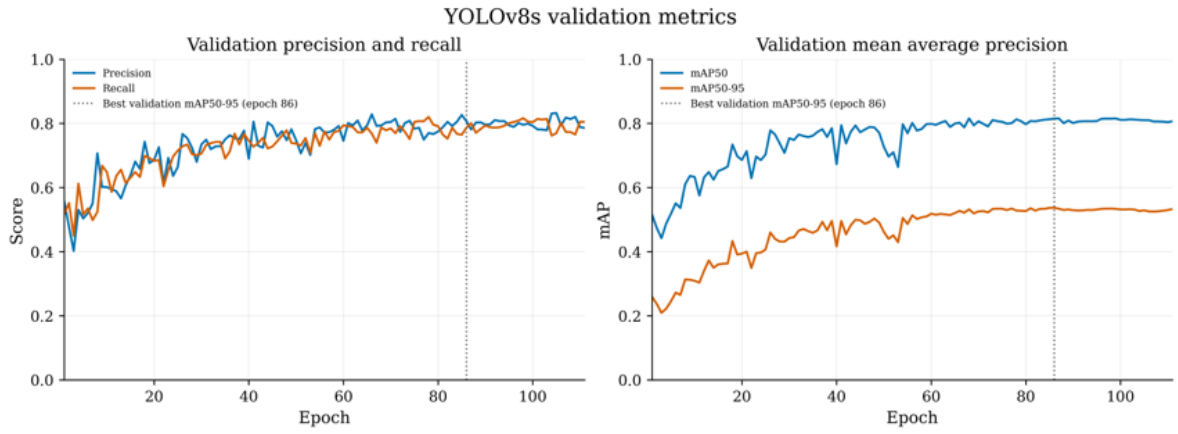



**Figure 7.1 YOLOv8s training and validation histories**

ViT-B/16 achieved 90.11% test accuracy. The best recorded validation loss was 0.2277 at epoch 3, with 92.29% validation accuracy. By epoch 13, training accuracy was 96.65%, validation accuracy 86.13% and validation loss 0.5096; early stopping was activated. This divergence supports concern about overfitting, but a repeated-run and external-farm evaluation is required to characterise it fully.

The segmentation baseline achieved approximately 0.298 mIoU and the refined model 0.4078 on 100 evaluated samples. The absolute improvement is 0.1098; the relative improvement is approximately 36.8% using the reported baseline. The refined overlap

remains moderate. Simultaneous changes to fine-tuning and balancing prevent attribution of the entire improvement to one change without an ablation experiment.

**Table 7.4 Smart-diagnosis experimental results**

| Stage or condition | Result | Evaluation scope |
|---|---|---|
| YOLOv8s selected checkpoint | Precision/recall ≈0.80; mAP@50 ≈0.81; mAP@50–95 ≈0.54 | Approximate validation-curve values, epoch 86 |
| ViT best validation | Loss 0.2277; accuracy 92.29% | Epoch 3; checkpoint selection context |
| ViT held-out test | Accuracy 90.11% | Aggregate four-class test result |
| Segmentation baseline | mIoU ≈0.298 | 100 evaluated samples |
| Segmentation refinement | mIoU 0.4078 | 100 evaluated samples; moderate remaining overlap |
| Segmentation improvement | +0.1098 absolute; ≈36.8% relative | Comparison using reported baseline |

Table 7.4 reports the updated component values. Per-class F1, calibrated confidence, controlled OOD rejection, cattle-to-buffalo transfer and full-cascade clinical accuracy are not established. The four-class classifier includes a cattle category, so the test result is not equivalent to a four-disease diagnostic sensitivity. Figure 6.3 combines the original classifier histories and segmentation comparison.

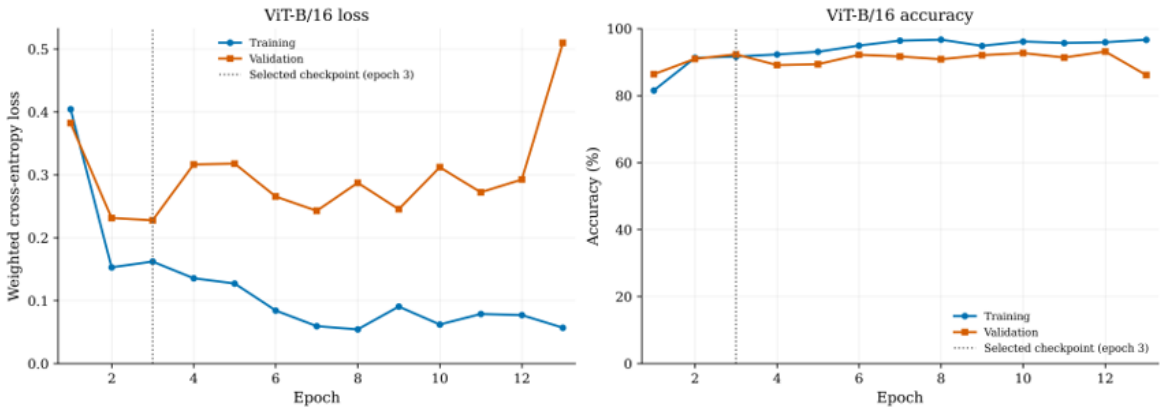



**Figure 7.2 ViT learning histories and segmentation refinement**

## 7.6 Geospatial results

The geospatial implementation provides historical origin selection, source inspection, relative cell scores, geometric descriptors and veterinarian-oriented case views. Its LSD reference calculation gives a reproducible distance-based ordering within the selected area of interest. The map and source panels support interpretation of the output, but interface functionality does not establish predictive accuracy of a biological transmission process.

Quantitative evaluation of FMD spatial ranking requires a disease-specific record identifying what was ranked, which outcomes were held out, the observation denominator and the baseline. A complete reproducible benchmark is not established for the spatial component. Its contribution is therefore supported by the defined calculation, operational interfaces and qualitative demonstration, with predictive evaluation remaining a priority.

The distinction between geometric direction and biological movement is particularly important. A negative-gradient direction follows the shape of the score field. Apparent historical rate depends on reported timing and location, which may include reporting delays. Nominal reach compounds those assumptions. Neither descriptor independently verifies future spread. Figure 6.4 presents a selected origin and sources, retaining the retrospective labels.

*FIGURE 6.4 PLACEHOLDER*

*Insert a geospatial analysis screenshot showing selected historical origin, source markers, approximately 5 km cells, relative score legend and geometric direction/reach context. Label the example as retrospective and avoid probability terminology for the cell score.*

**Figure 6.4: Retrospective spatial score and source inspection example**

Source: Authors' own architecture or application evidence.

## 7.7 District-month outbreak-risk results

FMD results are reported separately for the default 30-feature model and the enhanced 31-feature variant. Table 7.5 shows the walk-forward means and held-out 2024 discrimination. The same-district previous-month feature improved ranking in the recorded experiment, but the enhanced service depends on that status being available and verified. The association is predictive rather than evidence of causal transmission.

**Table 7.5 FMD discrimination results by model and evaluation**

| Configuration | Evaluation | ROC-AUC | PR-AUC |
|---|---|---|---|
| 30-feature baseline | Walk-forward mean 2022–2024 | 0.7833 | 0.3773 |
| 31-feature history variant | Walk-forward mean 2022–2024 | 0.8120 | 0.4698 |
| 30-feature baseline | Held-out 2024 | 0.7570 | 0.5461 |
| 31-feature history variant | Held-out 2024 | 0.8166 | 0.6968 |

The held-out 2024 grid contains 300 district-months, including 78 outbreak-positive observations. At the evaluated threshold of 0.40, the baseline identifies 51 of 78 positives and the enhanced variant 53 of 78. Table 7.6 reports the threshold trade-off. Appendix D includes the complete confusion counts, making the precision and recall denominators explicit.

**Table 7.6 FMD threshold-specific held-out 2024 results at 0.40**

| Configuration | Recall | Precision | F1 | Outbreaks detected |
|---|---|---|---|---|
| 30-feature baseline | 65.4% | 50.5% | 0.570 | 51/78 |
| 31-feature history variant | 67.9% | 53.0% | 0.596 | 53/78 |

The enhanced model still misses 25 outbreak-positive district-months and raises 47 false-positive alerts in this test. Stronger AUC therefore does not establish sufficient performance for automatic veterinary intervention. The threshold of 0.40 concerns outbreak classification and conditional Stage 2 evaluation; it should not be confused with every LOW, MEDIUM and HIGH presentation-band boundary.

LSD performance is more modest. Table 7.7 traces the recorded model progression under five LOYO folds across 2020–2024. The final Platt-calibrated ElasticNet achieves mean ROC-AUC 0.6136 and PR-AUC 0.2699. Although calibration is intended to improve probability interpretation, these discrimination metrics alone do not quantify final calibration error.

**Table 7.7 model progression under five-fold LOYO evaluation**

| Configuration | Mean ROC-AUC | Mean PR-AUC |
|---|---|---|
| 23-feature climate baseline | 0.5308 | 0.1276 |
| 27-feature spatial model | 0.5559 | 0.1657 |
| 28-feature history model | 0.6034 | 0.1876 |
| 28-feature ElasticNet | 0.6167 | 0.2689 |
| 28-feature Platt-calibrated ElasticNet | 0.6136 | 0.2699 |

The shorter, more imbalanced LSD history and between-year variation limit confidence in consistent generalisation. LOYO keeps complete years separate, but it permits later observations in training for some folds. The result should therefore be described as year-separated out-of-sample discrimination. Prospective forecasting requires strictly forward validation and verification of which features are available at the forecast issue date.

Offline FMD Mondrian conformal analysis reports approximately 95.3% overall empirical coverage and 94.9% outbreak-class coverage, with around 41% single-class sets. The remaining sets generally contain both classes. This shows an evaluated uncertainty approach, not deployment of full conformal sets through every runtime endpoint. Temporal dependence and distribution shift must be considered before interpreting coverage as a prospective guarantee.

Figure 7.3 compares the FMD baseline/variant and LSD progression using the recorded values, keeping their evaluation protocols visibly separate. Current-month features and substituted historical conditions also mean that the application supports risk nowcasting and climatological projections; it has not established validated long-range future-weather prediction.

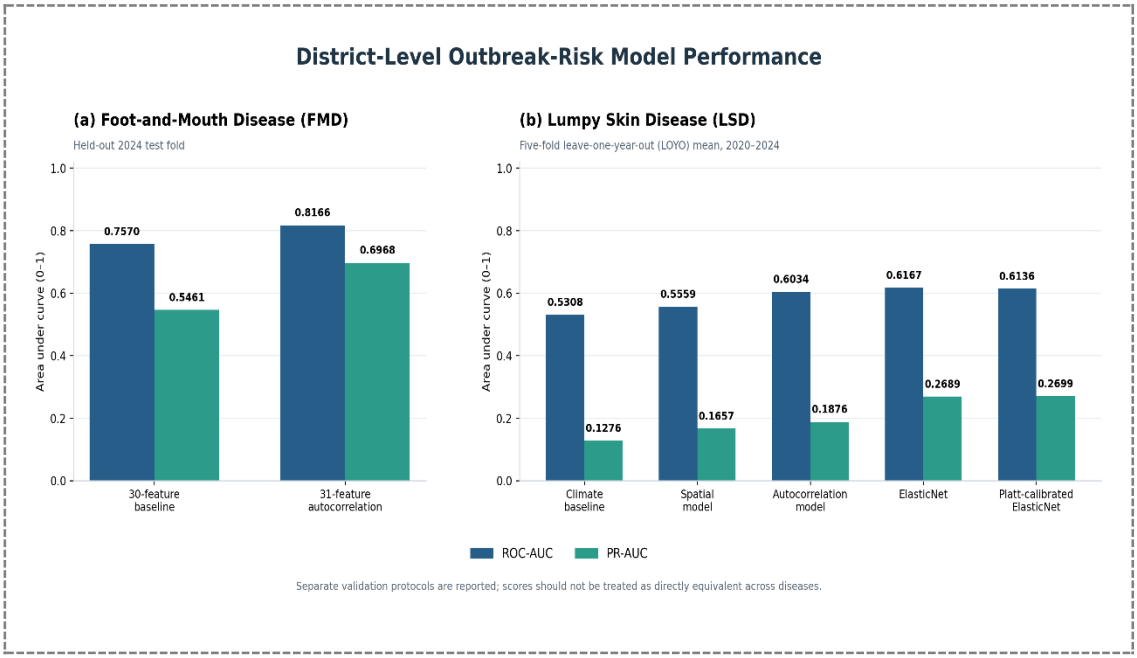



**Figure 7.3 Comparison of FMD (2024 held-out) and LSD (five-fold LOYO) model performance using ROC-AUC and PR-AUC.**

## 7.8 Interpretation of the integrated evidence

The components provide complementary evidence rather than one combined performance score. Wellness tests support the constructed triage logic; image experiments support stage-level visual recognition; geospatial views support source-aware spatial interpretation; and forecasting provides disease-specific district-month discrimination and implemented operational services. None alone demonstrates the performance of the entire veterinary workflow.

The strongest general conclusion is feasibility of the information architecture and several useful component artefacts. The most important limits are simulated triage validation, moderate segmentation overlap, missing image-service handover, incomplete spatial predictive evidence, modest LSD generalisation and the absence of a prospective whole-system study. These limits guide prioritisation of further work rather than invalidating the narrower experiments.

## 7.9 Stakeholder and User Validation

The triage thesis reports qualitative feedback from field veterinarians about the dual-trigger workflow and management guidance. It does not provide a participant count,

structured instrument, recruitment details or quantitative analysis sufficient to establish representative acceptance. The feedback is retained as reported design input, without inferring reductions in antibiotic use or improved clinical outcomes.

The diagnosis thesis does not establish a completed veterinary-panel or farmer-usability study. The forecasting thesis presents role-oriented functional validation, while explicitly distinguishing it from formal DAPH or veterinary user acceptance. Geospatial demonstrations show how maps and district views communicate information, but screenshots and demonstrations do not establish decision quality in practice.

A full validation study should recruit representative farmers, veterinary officers and surveillance users, obtain the required approvals and consent, and record tasks and assessment criteria in advance. It should assess comprehension of suspected versus confirmed cases, Heat Stress versus Unhealthy labels, relative spatial scores versus probabilities, and historical proxies versus observed climate. Interviews can identify workflow concerns, while task observations and decision review provide measurable outcomes.

Field evaluation should also measure reporting timeliness, completeness, false alerts, review burden, unresolved cases and usefulness of investigation prioritisation. Veterinary reference standards and laboratory confirmation should be defined for disease-specific claims. Adoption and economic benefits require evidence over sustained use, not only a one-session interface demonstration.

## 7.10 Achievement of Common Objectives

Table 7.8 maps evidence to the six common objectives. Partial achievement reflects either a remaining integration boundary or a broader validation requirement; it does not mean that the narrower recorded experiment failed. The report avoids a full achievement claim where the objective includes field or system-level behaviour that has not been measured.

**Table 7.8 Achievement of common objectives**

| Objective | Main evidence | Status and justification |
|---|---|---|
| GO1 Wellness and anomaly triage | Engineered-data accuracy, simulated comparison and service flow | Partly achieved; clinical lead time and BCS error remain unvalidated |
| GO2 Image assessment pipeline | Detector curves, 90.11% ViT test accuracy and improved mIoU | Partly achieved; stage target met, runtime cascade and field transfer pending |
| GO3 Spatial interpretation | Defined reference score, case adapters and national/district views | Partly achieved; biological direction and future spread not established |
| GO4 District-month risk | FMD walk-forward, LSD LOYO and variant/proxy-aware APIs | Partly achieved; implemented and evaluated, prospective validation limited |
| GO5 Common workflow integration | Shared services and selected live operational workflows | Partly achieved; all-component confirmation chain not demonstrated |
| GO6 Whole-system validation | Recorded automated checkpoint and documented limits | Partly achieved; field acceptance, load, recovery and security audit pending |

## 7.11 Integrated Limitations and Threats

The main construct-validity threat is conflating different outcomes. Unhealthy wellness, image candidate disease, verified clinical event, historical source, relative cell score and district-month occurrence probability are distinct constructs. Combining

them as confirmed disease or using one component's score as another component's ground truth would invalidate the interpretation of results.

The triage experiments depend on engineered rules, breed baselines and reconstructed sequences. A random split can preserve much of the scenario-generating structure in both training and test. The 20-profile comparison is small and may have influenced model selection. Independent animal-level and farm-level longitudinal evaluation is required to measure clinical sensitivity, false-alert burden and lead time.

Image validity is limited by heterogeneous source datasets, label remapping, class imbalance and incomplete split lineage. Multiple crops or near-duplicate images from the same animal can create dependence unless grouping is enforced. Annotation-guided classifier crops differ from actual detector crops. Detector estimates are approximate, segmentation has moderate overlap, and the absence of external farm/species, per-class and calibration results limits deployment conclusions.

Spatial analysis is affected by coordinate uncertainty, reporting delays, source duplication and historical eligibility rules. A frozen distance kernel supports consistent ranking, not a direct disease-transport mechanism. Geometric direction and apparent rate may be sensitive to the observed source set. Comparisons with future outbreak observations require origin-specific, time-bounded evaluation and a suitable baseline.

Forecasting validity is limited by sparse positive observations, reporting heterogeneity and imperfect provenance certification for some reconstructed records. Complete zero-filled grids may confuse unreported events with verified absence. Environmental products describe broader conditions rather than farm microclimate. Previous-month status and current-period inputs must be checked against their actual availability at issuance; otherwise an apparently strong nowcast can be mistaken for an advance forecast.

FMD walk-forward validation is closer to chronological use than LSD LOYO, but neither automatically establishes performance under a changed reporting or climate regime. Future proxy selection uses historical summary information and is not a measured future-weather model. Risk explanations describe fitted associations, not

causal drivers. Conformal coverage is offline and depends on assumptions that require reassessment under temporal change.

At system level, component dependency configurations and release states are not fully harmonised. Contradictory infrastructure descriptions, missing image orchestration and checkpoint-specific tests limit reproducibility. Selected route protections do not establish complete authorisation. No full-platform latency, availability, restoration, penetration, accessibility or prospective stakeholder study is established. Consequently, production readiness and public-health or economic benefit remain open evaluation questions.

# 8. COMMERCIALISATION, SUSTAINABILITY AND RESPONSIBLE INNOVATION

## 8.1 Value Proposition and Beneficiaries

ADRS provides a common place to record animal concerns, examine AI-supported evidence, review veterinary cases and consult spatial or district-level surveillance information. Its proposed value lies in reducing disconnected records and making the origin and interpretation of evidence visible. The application does not need to guarantee diagnosis or disease prevention to provide a useful organisational service.

Farmers are potential beneficiaries of structured observations and clearer access to review. Veterinary officers can benefit from a consolidated history, supporting images and district context. District and national authorities can use verified events and model estimates to support investigation planning. These are intended benefits, not measured increases in productivity, reductions in losses or institutional commitments.

The commercial position should be based on the integrated workflow and local fit rather than unsupported superiority of individual models. Real adoption depends on useful outputs, acceptable review burden, affordable operation and reliable data. Lower hardware requirements for ordinary image capture may help access, but cloud execution, connectivity and maintenance remain recurring costs.

## 8.2 Market and Competitive Position

The initial target market is institution-supported veterinary surveillance and participating livestock farms in Sri Lanka. Government veterinary offices, agricultural organisations and authorised service providers are plausible adopters. The report does not establish a procurement agreement, addressable-market estimate, customer willingness to pay or signed deployment partnership.

Relevant alternatives include DAPH surveillance processes, FAO reporting and disease-intelligence tools, WOAH WAHIS, general farm-record software and task-specific precision livestock products. FAO EMA-i already supports field disease-event reporting [6], and WAHIS provides internationally reported animal-health information

[7]. ADRS should be positioned as a complementary local decision-support workflow rather than as the first digital reporting solution.

Table 8.1 gives a functional comparison without asserting price or performance superiority. A formal assessment would require matched tasks, local user studies, total operating cost and the licensing conditions of each alternative. Published image accuracies from different datasets are not a commercial benchmark.

**Table 8.1 Functional positioning against relevant approaches**

| Approach | Principal strength | ADRS contribution and constraint |
|---|---|---|
| Existing veterinary reporting | Institutional responsibility and clinical review | Adds connected evidence views; requires workflow agreement |
| International event platforms | Broad disease reporting and intelligence | Adds animal/farm context; not a replacement for official notification |
| Sensor-based farm monitoring | Frequent measurements in equipped farms | Uses logs/images; depends on user input and validation |
| Standalone image classifiers | Focused visual recognition | Connects stage evidence to review; full cascade pending |
| Spatial or district risk models | Population-level interpretation | Presents both perspectives with distinct definitions |

## 8.3 Business Model and Feasibility

An institution-supported service is a reasonable initial sustainability model because surveillance has shared benefits and requires professional oversight. A limited pilot could involve selected veterinary offices and consenting farms, then expand according to usability, data quality, measured operating cost and demonstrated workflow value. This is a proposed adoption strategy rather than an established revenue plan.

Possible funding arrangements include institutional licensing, implementation grants, support and maintenance contracts, or partnerships with agricultural and veterinary organisations. The preferred arrangement should be decided after measuring recurring hosting, database, model execution, data updates, user support and training costs. Student cloud credits can support experimentation but are not a durable operating model.

Technical feasibility is supported by the common web stack, implemented forecasting services, geospatial interfaces and research artefacts. Commercial feasibility is less established. The image runtime boundary, data licensing, model maintenance, production security and institutional acceptance require completion. Costs should be reported per supported farm, review workflow or service volume rather than inferred from one short inference benchmark.

A staged programme should first stabilise the release and validate the clinical confirmation flow, then conduct a limited supervised pilot. Expansion should be conditional on acceptable false-alert burden, reporting completeness, reviewer workload, service reliability and total cost. Component outputs that do not improve decisions should be revised or omitted rather than retained only because a model was trained.

## 8.4 Adoption, Intellectual Property and Compliance

Technology adoption depends on digital literacy, trust, connectivity, support and compatibility with farm routines. The Italian adoption findings provide useful context for cost and interpretation barriers [8], but local assessment must determine whether the same barriers apply. Mobile-friendly forms, clear labels, assisted onboarding and short veterinary training materials can be evaluated during pilot use.

Farm images, locations, identities and clinical records require governance before broader deployment. A deployment should define collection purpose, access, retention, correction and deletion processes, together with responsibility for operational data. The applicable Sri Lankan data-protection framework includes the 2022 Act and 2025 amendment [27]. Legal compliance and institutional approval must be assessed for the actual processing arrangement; a report describing encryption or anonymisation does not itself establish compliance.

The triage thesis reports consent and anonymisation practices, whereas other component studies use secondary research data. A group-level ethics account must distinguish these cases and retain actual permissions and approvals. Precise farm coordinates can identify a household even after removing a name, so analytical maps and source exports require appropriate scoping. Formal ethics approval and security certification remain outside the established evidence.

Intellectual property review must cover software ownership, team contribution, university policy, dataset permissions and pretrained-model licences. Ultralytics identifies AGPL-3.0 and enterprise licensing options for its software [18] . The applicability to the selected distribution and hosted-service arrangement requires review before commercial release. Publicly accessible images or surveillance pages must not be assumed to permit unrestricted redistribution.

Responsible innovation requires clear communication of uncertainty and professional accountability. A model candidate does not authorise treatment, and a district estimate does not authorise movement restrictions. Heatmaps and feature contributions can assist explanation but do not certify clinical truth. The confirmation boundary, model

version and provenance should remain visible so that a reviewer can correct an assessment and retain the reason for that decision.

## 8.5 Sustainable Development Goal Contribution

The prospective SDG 2 contribution is improved information for resilient livestock production and small-scale producers [9]. The prospective SDG 3 connection is improved animal-health information relevant to wider preparedness [10]. The project does not demonstrate increased food output, reduced human disease, lower mortality or reduced antibiotic use.

Meaningful impact evaluation should measure whether reporting becomes faster or more complete, whether veterinarians find the information useful, and whether follow-ups reach farms that need assessment. Productivity or disease-control outcomes require a longer design with suitable comparators and control for changing management conditions. Adoption rates and geographic coverage should also be examined to avoid benefits being concentrated in farms with better connectivity.

Sustainability requires ongoing model maintenance, surveillance updates, secure operation and an affordable support arrangement. Unreliable data, excessive false alerts or opaque provenance can undermine the intended value. The platform's contribution to the goals should therefore be treated as a hypothesis for institutional evaluation rather than an impact already established by the research.

# 9. CONCLUSION AND FUTURE DIRECTIONS

## 9.1 Overall Conclusion

This research developed a common framework for connecting animal wellness observations, image-based assessment, geospatial surveillance and district-month outbreak-risk information. ADRS provides a veterinary-supervised workflow in which these different sources can be recorded and interpreted together. Its principal result is a coherent information and service architecture supported by component artefacts and selected operational integration evidence.

The wellness work reports high agreement on engineered data and a small simulated triage batch, while leaving clinical generalisation and BCS regression error open. The image work reports approximate YOLO validation metrics, 90.11% ViT test accuracy and improved segmentation mIoU of 0.4078. The spatial work provides historical-source-aware interpretation and verified-case views, with relative scores and geometric descriptors kept distinct from biological transmission claims.

The district-month work provides stronger FMD than LSD discrimination. The enhanced FMD model achieves held-out 2024 ROC-AUC 0.8166 and PR-AUC 0.6968; the calibrated LSD model achieves five-fold LOYO mean ROC-AUC 0.6136 and PR-AUC 0.2699. The forecasting services expose variant and proxy handling, and a recorded automated checkpoint reports 862 passes. These results support the narrower experiments and workflows but do not establish full-platform clinical accuracy or production readiness.

All common objectives are partly achieved when judged against their complete intended scope. The outstanding image-service handover, field validation, security and recovery checks prevent a claim that the complete system has been validated end to end. ADRS therefore provides a practical research foundation for integrated livestock-health decision support, with clearly identified work needed before wider veterinary operation.

## 9.2 Principal Integrated Contribution

The contribution is the connection of evidence at different levels: animal-level daily observations and body condition, image-level candidate findings, event-level veterinary confirmation, cell-level historical spatial ranking and district-month occurrence probability. Shared identities, defined contracts and explicit verification status allow these outputs to coexist without being mistaken for interchangeable clinical measurements.

The project applies established software, computer-vision, multimodal and tabular learning techniques to a common Sri Lankan livestock workflow. Novelty is concentrated in the integration, local data preparation, component adaptations and responsible presentation of evidence. The research does not claim a new YOLO, ViT or ResNet architecture, universally superior prediction, or a demonstrated replacement for veterinary services.

The design also makes uncertainty operationally visible. Historical proxies, model variants, limited clinical labels, retrospective spatial origins and optional masks each have a defined interpretation. Maintaining those distinctions is central to the shared system's value, because useful decision support depends on understanding what an output represents as well as its numerical result.

## 9.3 Future Directions

The priority is to stabilise one reproducible integrated release. This requires compatible dependencies, configured model paths, frozen artefact versions, verified detector/classifier mappings and a working diagnosis service. A full regression should trace one concern through triage, image analysis, review, confirmation, map update, forecast consultation and follow-up, including corrected and rejected cases.

The second priority is independent veterinary validation. Wellness assessment requires real longitudinal animal records, an external farm split, numerical BCS MAE/RMSE and comparison against expert assessment. Image assessment requires per-class results, exact detector exports, farm/animal grouping, actual predicted crops, poor-

quality cases and unfamiliar-species or OOD tests. Calibration and referral thresholds should be evaluated before confidence scores are used operationally.

The third priority is stronger spatial and temporal evaluation. Spatial ranking should be tested against time-bounded later outcomes with transparent origins, denominators and baselines. Direction and reach require comparison with independently observed movement or spread before biological claims can be made. District models should use strict forward validation, feature-availability cutoffs, calibrated-probability assessment and additional surveillance years, especially for LSD. Conditional severity needs its own reference standard and evaluation.

The fourth priority is production engineering and governance. This includes consistent backend role/district authorisation, file and request controls, secret management, dependency checks, worker-memory profiling, load tests, dependency-failure recovery and tested backup restoration. The active weather/database configuration, dataset licences and applicable ethics and privacy arrangements must be resolved and documented.

The fifth priority is supervised adoption research. Representative farmers, veterinary officers and institutional users should assess task completion, comprehension, false-alert burden, reporting timeliness and usefulness of follow-up prioritisation. A live notification provider should be tested only within an approved operating workflow. Economic and SDG-related claims should follow measured sustained outcomes rather than component accuracy alone.

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## 11. GLOSSARY

Body condition score: an assessment of an animal's body reserves. In ADRS it is estimated from images, with expert agreement still requiring quantitative validation.

Calibration: agreement between predicted probabilities and observed outcome frequencies. Discrimination metrics alone do not establish calibration.

Confirmed case: an operational clinical record verified through the authorised veterinary workflow. It is distinct from an AI candidate or historical source event.

Historical proxy: past conditions substituted for unavailable exact-period inputs. It is not measured weather for a future month.

Late fusion: combination of representations derived from separate input branches before a final prediction.

Nowcasting: estimation of a current or near-current condition using information available for that period. It differs from forecasting strictly ahead of all target-period observations.

Relative spatial score: an ordering of cells under a specified source and calculation. It is not automatically a calibrated infection probability.

Region of interest: a selected image area passed to downstream processing. Its identity and coordinate transformation must be retained.

Veterinary decision support: information supplied to assist professional assessment and planning, with responsibility for clinical and control decisions remaining with authorised personnel.

# 12. APPENDICES

The appendices consolidate the interface, contribution, testing and supplementary result material referenced in Chapters 4–6. They preserve evidence definitions and identify review requirements without substituting for the main explanation.

### Appendix A Integrated Interface Specification

The specification in Table 12.1 records implemented interface shapes and the required smart-diagnosis handover. Fields described as required contract elements express the integration specification; they do not imply that every endpoint currently returns them under identical names. The final release should preserve a versioned schema and verify it against frontend consumers.

**Table 12.1 Cross-component record and payload specification**

| Record or payload | Required content | Validation and interpretation |
|---|---|---|
| Daily observation | Animal/farm identity, observation date, milk yield and body weight | Nonnegative units, authorised owner and no duplicate day |
| Seven-day triage | Ordered 7 × 8 sequence, 45-feature static vector and BCS | Exact feature order; missing/stale weather flagged |
| Image analysis handover | Image/ROI identity, boxes, canonical labels, scores and model versions | Required service boundary; final runtime mapping pending |
| Verified clinical event | Case ID, disease, event date, location and verifying actor | Distinct from prediction; scoped clinical access |

| | | |
|---|---|---|
| Historical spatial event | Source ID, disease/date, coordinates and provenance | Historical/research identity retained; deduplication recorded |
| Spatial snapshot | Origin/version identity, cells, sources and descriptors | Coherent chunks; score is relative; WGS84 order checked |
| District risk request | Disease, district, year, month and applicable variant | Supported domain; issue-time feature availability checked |
| District risk response | Occurrence probability, variant, explanation and provenance | Historical proxy and conditional Stage 2 identified |
| Operational follow-up | Forecast/case link, officer assignment, state and audit time | Role authorisation, update conflict and retry controls |

Observation date, event date, review timestamp and analysis issue time must not be used interchangeably. Animal IDs and clinical case IDs are operational identifiers; historical event IDs belong to a source lineage. District names require a canonical registry or explicit normalisation to prevent spelling variants from changing feature lookup or access scope.

Probability validation should check finite values and range. The four-class score vector requires a stable class-name mapping and consistent length; it must not be reinterpreted as multi-label coexistence. Spatial cells should retain geometry and selected-origin identity. A failed service must not return an empty array that the client interprets as a successful absence of disease.

## 12.1 Appendix B Team Contribution Declaration

The contribution allocation in Table 5.5 forms the basis of this declaration. Each member is responsible for reviewing the accuracy of the recorded individual work and the shared report. Signatures certify the authors' own agreement; they are intentionally left for completion by the team.

**Table 12.2 Team contribution review and sign-off**

| Member and student ID | Primary component | Signature | Date |
|---|---|---|---|
| A.L.M. Athulathmudali — IT21129544 | AI Smart Diagnosis | _____ | _____ |
| W.M.P.J. Wijenayake — IT22194558 | Persistent Health Anomaly and BCS | _____ | _____ |
| G.A. Sandaru — IT22258908 | Geospatial Clustering and Trajectory Mapping | _____ | _____ |
| S.S. Kumarasinghe — IT22221414 | Climate-Informed Livestock Disease Forecasting |  | 2026.10.11 |

Shared responsibilities include the common problem definition, integration contracts, application navigation and identity context, synthesis of evidence and final report review. A member's signature does not establish independent clinical validation of every model; the evaluation boundaries described in Chapter 6 remain applicable.

## 12.2 Appendix C Integrated Acceptance and Evidence Matrix

Table 12.3 summarises acceptance cases for the complete release. Its status distinguishes reported component evidence from verification still needed. Expected behaviours are test criteria, not newly executed results.

**Table 12.3 Complete-workflow acceptance cases**

| Test case | Expected behaviour | Evidence status |
|---|---|---|
| Invalid or zero baseline | No division exception; missing context remains visible | Triage guard reported; missingness review needed |
| Incomplete seven-day payload | Reject or explicitly identify unavailable observations | Shape guards reported; full recovery pending |
| Weather timeout | Mark unavailable source; avoid silent normal prediction | Dependency-failure acceptance pending |
| Known image full cascade | Preserve box, ROI, class mapping and mask identity | Stage artefacts available; complete test pending |
| Unfamiliar/poor-quality image | Escalation or recapture without confirmed healthy claim | OOD and field robustness pending |
| Suspected case confirmation | Only authorised professional creates verified event | Host adapters described; complete chain pending |
| Cross-user/district request | Backend denies unauthorised record access | Selected role tests; release-wide audit pending |
| Stale spatial snapshot | Reject mismatched origin or version | Contract checks described |

| | | |
|---|---|---|
| No exact district-month row | Identify proxy and suitable model fallback | Forecasting behaviour and tests reported |
| Missing model artefact | Explicit unavailable status without fabricated risk | Forecasting error handling described |
| Forecast-to-follow-up workflow | Retain record link, assignment and audit identity | Live HTTP checkpoint reported |
| Notification request | Draft/simulated result labelled; no false delivery claim | Simulated provider described |
| Concurrent complete workflows | Bounded latency and stable memory without lost records | Full-platform load test pending |
| Backup restoration | Restore coherent records and matching artefact versions | Whole-system restore test pending |

Each execution record should identify commit, date, environment, inputs, expected output, observed output and pass/fail result. Forecasting's recorded 862 passing tests must not be added to unnamed overlapping suites or applied to a later release without rerunning the relevant checks.

## 12.3 Appendix D Supplementary Forecasting Results

Table 12.4 gives the held-out 2024 FMD confusion counts at threshold 0.40. Both configurations use 222 negative and 78 positive district-months. These counts explain the threshold-specific results in Table 6.6 and preserve the distinction between ranking measures and operational alert performance.

**Table 12.4 FMD held-out 2024 confusion counts at threshold 0.40**

| Configuration | True negatives | False positives | False negatives | True positives |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 30-feature baseline | 172 | 50 | 27 | 51 |
| 31-feature history variant | 175 | 47 | 25 | 53 |

For the baseline, precision is 51/(51 + 50) and recall is 51/(51 + 27). For the enhanced variant, precision is 53/(53 + 47) and recall is 53/(53 + 25). Rounded results are 50.5% and 65.4% for the baseline, and 53.0% and 67.9% for the enhanced variant. The model choice must also consider the availability of verified previous-month status.

Table 12.5 summarises the runtime model roles. Stage 1 estimates occurrence; Stage 2 provides conditional advisory information. LOW, MEDIUM and HIGH labels from different stages or components must not be assumed to have the same meaning.

**Table 12.5 Forecasting model configurations and operational roles**

| Disease or stage | Model configuration | Operational interpretation |
|---|---|---|
| FMD Stage 1 | 30-feature Logistic Regression | Default baseline without required own-district lag |
| FMD enhanced Stage 1 | 31-feature Logistic Regression | Uses verified previous-month occurrence |
| FMD Stage 2 | Random Forest | Conditional advisory severity |
| LSD Stage 1 | 28-feature Platt-calibrated ElasticNet | Primary occurrence model with own-district lag |
| LSD fallback Stage 1 | 27-feature ElasticNet | Used when previous-month status is unavailable |
| LSD Stage 2 | Logistic Regression | Primarily quiet-period suppression; active severity limited |

## 12.4 Appendix E Academic Integrity and AI Assistance Disclosure

Generative AI assistance was used to help consolidate the group report, organise the component summaries, identify inconsistencies, locate external references and prepare document formatting and figure placeholders. The analysis relies on the project documents and reported experiments; no model retraining, new field trial or new software test execution is claimed by this report preparation.

The authors retain responsibility for checking source accuracy, dataset permissions, numerical results, contribution statements and compliance with the applicable module policy. AI assistance does not establish consent, institutional approval, implementation completeness or clinical correctness. The team should review the completed wording, confirm the relevant model and release artefacts, and provide any additional disclosure required by the institution.